The Price of Corruption: Evidence from Lava Jato and the Construction Club in Peru

Carlos Chavez*

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Abstract

Corrupt public works are rarely identified, so their cost is hard to measure. Two Peruvian legal records identify them: the Odebrecht bribery case and a 2021 ruling against thirty-two construction groups for fixing road contract prices. Projects with a documented bribe went over budget by 53 percent, the rest by 17. Where firms fixed the price the winner charged 108 percent of the government's cost estimate, against 90 percent when they competed, and the same firm's bid was about four points higher against that estimate on a contract it had fixed. Households next to a corrupt road did worse than those next to a comparable road where the firms competed, though across a wider forty-kilometre area we cannot detect a difference. The ruling also lets us test the methods used to catch price fixing. Ranking contracts by the winning price puts a fixed one above a competitively bid one 87 percent of the time. Ranking them by how far apart the bids are does no better than chance.

JEL classification: D73, H57, K42, L74, D44, O18.

Keywords: corruption, public procurement, bid rigging, collusion detection, road infrastructure, household welfare, Peru.

1 Introduction

Corruption in infrastructure is hard to study because the corrupt projects are rarely identified. Peru offers two legal records that identify them. The first is the Odebrecht (Lava Jato) case, whose plea agreements and prosecutions name the road concessions, the bribes paid, and the officials who took them. The second is Resolución 080-2021/CLC, in which the Peruvian competition authority sanctioned thirty-two construction groups for fourteen years of price fixing in national road contracts. Lava Jato is bribery to win large concessions. The Club is firms agreeing in advance which of them would win each ordinary contract. Together they let us ask two questions with the corruption already established by the rulings. Did the projects it touched cost the State more, and were the places they were built in any better off?

The paper makes two substantive contributions and one methodological one.

The first is the price the State paid on the projects these records name. Of the twenty-five Odebrecht projects in Peru, the ten with a documented bribe went over budget by 53 percent on average. The fifteen others went over by 17. That difference survives the checks a sample of twenty-five allows, though it weakens once controls are added. On the cartel side we rebuild the

*Center for the Economics of Human Development, Department of Economics, University of Chicago. Email: cchavezpadilla@uchicago.edu. All errors are my own.

Construction Club from the ruling into seventy-eight road contracts and the firms that won or covered each one. No firm dominates. The wins are spread out, taken in turns and often shared between firms bidding together. The gain shows up in the price. Peruvian road contracts must be bid within a band around a cost estimate the government publishes, so a cartel does not need to bid high, only to stop discounting. Where the firms had agreed the winner beforehand the winning bid was 108 percent of that estimate. Where they competed it was 90.

The second is who bore the cost locally. The concession corridors show no household gain we can detect. Districts near cartel contracts do worse on average, but so do districts near road contracts that were competitively bid, so at that level of detail the pattern belongs to large road projects and not to corruption. To separate the two we pair each cartel contract with a comparable competitively bid one and compare the districts around each. Households around a cartel road have lower income and consumption and more poverty. In the years before the contract the two groups mostly track each other, with one gap four years ahead of the tender that shows up in consumption and in poverty. Under stricter pairing rules the estimates grow, and the income gap widens from about 14 to about 17 percent.

That number compares the districts immediately around a cartel road with those around its paired competitive road, not the wider region. Widening the area to forty kilometres and treating it as a single unit, the same comparison gives -0.008 on income and -0.004 on consumption, indistinguishable from zero however the districts are weighted. That rules out a regional loss as large as the local one and does not rule out a smaller one. Within the affected radius the loss falls mainly on earnings, and mostly among the self-employed. Fewer people are working as well, by a smaller amount that the data measure less precisely. It begins within two years of the contract being awarded, which a problem with the finished road would not explain. Ten competitive roads can be located, eight of them find a match and six survive the stricter rule, and that small number limits the precision of everything in this section.

The third contribution follows from the ruling itself. Competition authorities and researchers use statistical tests to flag contracts that may have been fixed. Those tests have been checked against known cartels before (Fazekas et al., 2026), though usually case by case. This ruling names the fixed contracts one at a time and we hold their bids, so the tests can be graded on the same contracts whose corruption is established. A test based on how much the winner charged ranks a fixed contract above a competitively bid one 87 percent of the time. The other test in common use, based on how far apart the competing bids are, does no better than a coin toss.

The paper proceeds as follows. Section 2 describes the two judicial records and how they fit together. Section 3 places the paper in the literatures on corruption, public procurement, and collusion detection. Section 4 builds the data, reconstructing the cartel from the resolution and mapping it to districts. Section 5 sets out the empirical strategy. Section 6 reports what corruption did to the price the State paid, and closes by grading the tests used to detect price fixing. Section 7 asks whether the places these roads were built in were any better off. Section 8 asks where the loss falls and when it begins. Section 9 concludes.

2 Institutional background

Peru's road corruption is documented by two separate judicial processes that reach different layers of the same system. Figure 1 places them on a common timeline.

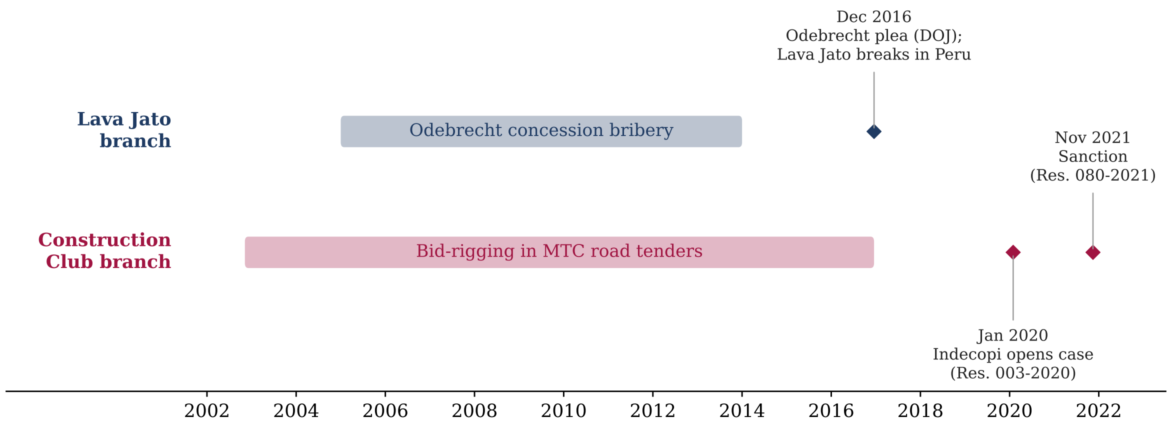

Figure 1: The two judicial branches of Peruvian road corruption.

Notes. Conduct periods (bars) and dated events (markers). Construction Club dates are from Resolución 080-2021/CLC (cartel conduct November 2002 to December 2016, opening of the procedure January 2020, sanction November 2021). Lava Jato dates are from the public record (Odebrecht bribery in Peru 2005 to 2014, the December 2016 plea agreement that broke the case).

The first window is the Odebrecht (Lava Jato) case. Odebrecht admitted to paying bribes to obtain public works in Peru over roughly a decade, and the December 2016 plea agreement with the United States Department of Justice, followed by the collaboration agreements with Peruvian prosecutors, named the projects and the officials. The corruption here operated mainly through concessions and megaprojects awarded at the top, the IIRSA corridors and urban works among them, and through the payments that secured them. The contracts were awarded mostly as concessions through the national investment-promotion agency, not through open procurement, and held by special-purpose vehicles financed by securitized debt and state co-financing. Odebrecht admitted roughly US$29 million in bribes in Peru, and four former officials, among them a former president, a former vice-minister of transport, and two regional governors, concentrate the documented payments. The revelation was an economic shock as well as a political one, with public-private infrastructure investment falling from around US$9 billion to under US$1 billion in 2016 and 2017 as projects froze. This is the branch an earlier version of this paper studied, and it identifies a set of road concessions as corrupt by judicial finding, not by inference.

The second window is the *Club de la Construcción*. In Resolución 080-2021/CLC of 15 November 2021 the Peruvian competition authority Indecopi sanctioned thirty-two construction groups for a horizontal bid-rigging agreement in the public procurement of national road works between November 2002 and December 2016. The mechanism was coordination among bidders. The firms agreed in advance on who would win each *licitación pública* run by the Ministry of Transport and Communications and arranged cover bids and abstentions to protect the designated winner. The procedure opened with the Resolución de Inicio 003-2020/ST-CLC of 31 January 2020. This branch reaches the procurement layer the concession cases do not, the ordinary road tenders, and it names the firms and their roles tender by tender.

The two records are complementary, not redundant. Lava Jato identifies the high-level bribery and concession channel but covers a small number of megaprojects. The Construction Club identifies the procurement-cartel channel across many ordinary road tenders and a wide geography. Studied together they document the same road-corruption ecosystem from the political allocation of concessions down to the firm-level rigging of tenders, and they let us measure the procurement-side cost and distortion of corruption against the local outcomes in the places the roads were meant to serve. The two channels also overlap in their firms. About twenty of the thirty-two sanctioned cartel groups appear as members of the Odebrecht concession consortia in the concession records, the ringed nodes in Appendix Figure 18, so the same construction oligopoly operated on both sides of the ecosystem, rigging the ordinary tenders below while partnering in the bribed megaprojects above. The two judicial records are therefore separate proceedings, not separate sets of actors. The household analysis tells the branches apart by geography, since the cartel tenders sit largely off the concession corridors, so the local comparisons identify off geographic variation. We keep the two as distinct records throughout and never merge them into a single database.

3 Literature review

The paper draws on four literatures. The first is the measurement of corruption, which is hard to observe and usually proxied by perceptions. Olken and Pande (2012) survey the move toward direct measurement, and Olken (2007) is its canonical road-sector instance, comparing official with engineer-estimated costs in Indonesian village projects. We measure in this direct tradition but from a different source, the judicial record. Campos et al. (2021) are the closest antecedent, reading cost overruns of bribed infrastructure off the Odebrecht case across nine countries, and we replicate their comparison on the Peruvian subset and read it as a magnitude. The line between a larger cost and a captured rent is the active-versus-passive waste distinction of Bandiera et al. (2009), and we describe the overruns as excess cost without attributing them to any party.

A second literature studies procurement award and performance. Its theoretical anchor for our setting is Burguet and Che (2004) and Compte et al. (2005), who show how corruption and competition interact inside an otherwise competitive procurement mechanism. Closest to us geographically, Lagunes (2021) runs a field experiment over two hundred Peruvian district governments and finds that monitoring public works lowers their cost without lowering the rate at which they are completed, which is the enforcement counterpart to the price distortion we measure. Decarolis (2014) shows the award price is not a sufficient statistic for value once performance and renegotiation are accounted for, which is why we treat the cartel through allocation and documented conduct instead of a single award figure. Coviello et al. (2018) show that discretion and repeated winners shape procurement outcomes, the pattern our reconstruction recovers, and Mironov and Zhuravskaya (2016) detect procurement corruption from firm financial transactions, a design whose data the Peruvian records cannot supply. Bosio et al. (2022) document across 187 countries that restricting procurement discretion helps low-capacity states and hurts high-capacity ones, and Peru is the low-capacity setting in which the discretion the cartel exploited is the object of interest.

A third literature, the one closest to our reconstruction, detects collusion in procurement auctions. Asker (2010) opens the internal organisation of a bidding cartel, the designated winner and the sharing of work over time that our reconstruction also finds, and Chassang and Ortner (2019) give the theory for collusion when bids are constrained, which is the environment a published reference value creates. Screens have been validated against known cartels before, most systematically by Fazekas et al. (2026) on a large sample of detected cases. Our addition is the unit at which the validation happens, since the ruling adjudicates tender by tender and we hold the bids of those tenders. Porter and Zona (1993) identify bid rigging in Long Island highway construction from the rank distribution of the bids, the canonical case of a ring that designates a winner while the others enter phony higher bids, which is the structure the cartel's exhibits record here. The modern wave screens for that conduct statistically. Conley and Decarolis (2016) recover bidder groups in collusive auctions, Kawai and Nakabayashi (2022) detect large-scale collusion among Japanese construction firms from rebidding, and Chassang et al. (2022) and Kawai et al. (2023) build robust screens and a regression-discontinuity test for noncompetitive bidding, with Pesendorfer (2000), Bajari and Ye (2003), and Marshall and Marx (2012) the foundations. Our setting inverts their inference problem. Where these designs must detect the designated-winner-and-cover-bid structure from bid data, the competition authority established it tender by tender, so we observe the assignment the screens have to infer. The closest structural match to our two-branch design is Clark et al. (2018), who study bid rigging and entry deterrence in Quebec construction through an investigation into collusion and corruption. We differ in reaching two layers of the same ecosystem through two separate judicial records, concession bribery above and tender rigging below, and in carrying both to household welfare, where procurement studies usually stop.

Brazil supplies the main Latin American benchmark for what corruption and its enforcement do to firms and workers. Szerman (2023) finds employment and earnings fall at firms debarred from public procurement, and Ferraz et al. (2025) trace the real effects of the Lava Jato investigation itself. Both study the enforcement shock rather than the corrupt procurement.

A fourth literature asks what transport infrastructure delivers locally. Donaldson and Hornbeck (2016) formalize market access as a sufficient statistic for the local gains, the construction our exposure measure uses, while the evidence on incidence is mixed: Faber (2014) finds China's highways concentrated activity in central cities, Ghani et al. (2016) find gains near India's Golden Quadrilateral, Morten and Oliveira (2024) trace roads through trade and migration to welfare, and Asher and Novosad (2020) find new rural roads in India raised neither income nor assets. Our no-dividend result for the concession corridors is consistent with that evidence, and Burgess et al. (2015), who show road building in Kenya followed political favoritism, is the premise our two records make concrete.

The paper differs from these literatures in its source. Where the designs above infer corruption from prices, bids or financial records, the ruling here names the tenders and the case record names the projects, so the price, the household contrast and the screen validation all rest on the same adjudicated set (Section 6.3).

4 Data

The judicial record of bribes and overruns. Treatment in Module 1 is a bribe documented in a court ruling, from the Odebrecht case dataset of Campos et al. (2021), restricted to the twenty-five Odebrecht projects in Peru. The dataset codes, per project, whether a bribe was paid, the bribe amount, and the cost overrun as a fraction of initial investment, with sector and initial and final investment. Ten projects carry a confirmed documented bribe, five are judicially implicated without a project-specific payment, and ten have no documented evidence. The corridors these projects built are mapped in Figure 2.

Figure 2: Odebrecht road concession corridors by documented-bribe status.
Notes. Documented-bribe corridors in warm shading (IIRSA Norte, IIRSA Sur T2 and T3) and the rest in cool. *Source.* MTC national road network (2017), bribe coding from Campos et al. (2021).

Road network and the accessibility measure. Module 2 is built on the MTC national road network for 2017 (*Red Vial Nacional*), which records each segment's surface. We node the network at intersections so that crossing roads connect at junctions, giving a graph of about 34,700 nodes whose giant component holds 99% of them. The network is sparse, as a national trunk network with few alternate routes in the Andes and Amazon is, and we report its connectivity. Travel time on each edge is its length divided by a surface-specific free-flow speed. District origins are the inhabited cells of the WorldPop 2017 population raster, covering the dispersed rural population and not only the district center, each carrying an explicit off-network feeder travel time to the nearest road node so that a remote settlement does not inherit a corridor node's access. Destinations are the 196 provincial economic centers, weighted by province population. The corridor interventions are coded by physical type, with the documented unpaved IIRSA routes paved (the conservative shock), the coastal Panamericana duplications and the Sierra rehabilitation treated separately, and corridor-specific pre-upgrade speeds, all documented with their basis and uncertainty. Table 1 collects the parameters.

Table 1: Market-access measure: parameters and corridor coding.

Element	Specification
Free-flow speed by surface (km/h)	
Rigid pavement / asphalt	80 / 70
Basic pavement / cobble	50 / 45
Gravel (afirmado / sin afirmar)	35 / 25
Earth track (trocha / emboquillado)	20–22
Off-network feeder	15
Gravity and geography	
Decay parameter θ	1 and 2
Origins	WorldPop 2017 inhabited cells, population-weighted
Destinations	196 provincial economic centers, population-weighted
Corridor upgrade coding (pre-upgrade speed)	
Conservative (IIRSA paving)	documented gravel routes at 22–30, paved post
Coastal Panamericana duplication	treated separately
Sierra rehabilitation	treated separately

Notes. Travel time on each network edge is its length divided by the surface-specific free-flow speed, with an explicit off-network feeder time from each populated origin cell to the nearest road node. The accessibility gain of a corridor differences market access before and after the documented upgrade, under the conservative and midpoint shock definitions. The Huánuco–Lima PE-3N defect is disclosed in Section 4. *Source.* MTC *Red Vial Nacional* 2017 and WorldPop 2017.

Route validation and a known defect. We validate the network against twenty known origin-destination routes across the coast, Andes, and Amazon. Eighteen are plausible against known travel times, one correctly remains unreachable by road (Iquitos), and one central-highway route, Huánuco to Lima, fails because of a break in the PE-3N corridor that forces an implausible detour. The defect is in the source network and is not yet repaired. Until it is we do not use the travel-time-to-Lima measure in any reported estimate, and the affected central-highland districts are flagged. The travel-time measure that does enter the results is time to the provincial capital, a different and generally shorter journey, and we have not traced which of those routes pass through the break. The market-access gradient, which differences the same network before and after the concession-corridor upgrades, is largely unaffected.

Household welfare. Module 3 uses the national household survey (ENAHO), *sumaria* module, 2004–2023, aggregated to a district–year panel of real per-capita household income and consumption and the poverty rate, deflated and person-weighted. ENAHO is designed to be representative above the district, so the district cells carry survey sampling noise, which the district and region-by-year fixed effects and the clustered inference are meant to accommodate. A district long-difference across the 2007 and 2017 population censuses provides the two-period check on population and composition that Appendix F reports.

Concession records. For the institutional note on concession continuity we use the OSITRAN performance reports, which show whether each road concession was rescinded or interrupted after 2016. We use these to establish that the contracts continued operating, not to infer the persistence of any rent.

Public investment. The fiscal-incidence test uses the MEF Invierte.pe project registry, public investment by district and function with initiation and completion dates, to test whether the cartel overcharge crowded out non-transport investment.

4.1 Descriptive statistics

Table 2 summarizes the household panel, which holds 21,938 district-year cells covering 1,692 districts over 2004 to 2023. The panel regressions run on 21,900 of these, because 38 districts are surveyed in a single year and drop out as singletons once district fixed effects are absorbed. Mean log per-capita income is 8.40 and mean log consumption 8.25, each with the dispersion across districts that a national panel spanning Lima and the rural Andes and Amazon would show, and the mean district poverty headcount is 0.42. Each cell pools a median of fourteen surveyed households, the sampling noise the district and region-by-year fixed effects and the clustered inference absorb.

Table 2: Summary statistics for the household panel.

	Mean	SD	Median	N
Panel A. District-year outcomes (2004–2023)				
Log household income	8.40	0.71	8.45	21,938
Log household consumption	8.25	0.61	8.29	21,938
Poverty rate	0.42	0.30	0.38	21,938
Households surveyed per cell	25.9	41.0	14.0	21,938
Panel B. Districts by exposure cell				
Neither corridor nor cartel				1,127
Lava Jato corridor only				165
Construction Club cartel only				332
Both branches				68
All districts				1,692

Notes. The household panel is the national household survey (ENAHO) collapsed to district-year cells, 21,938 cells across 1,692 districts for 2004 to 2023, of which the panel regressions use 21,900 because district fixed effects drop 38 singleton cells. Income and consumption are per capita and in logs, the poverty rate is the headcount, and the last row counts surveyed households underlying each cell. Panel B classifies a district by whether it is ever within ten kilometers of a Lava Jato concession corridor, of a Construction Club cartel tender, of both, or of neither. The cartel-only and both cells, 400 districts, are the ones the household analysis identifies from. *Source.* ENAHO *sumaria*, the Odebrecht concession corridors, and Resolución 080-2021/CLC.

Panel B places districts in the four exposure cells that organize the analysis. Of the 1,692 districts, 1,127 are near neither branch of the corruption, 165 lie near a Lava Jato concession corridor only, 332 near a Construction Club cartel tender only, and 68 near both. The household identification comes from the 400 districts ever exposed to the cartel procurement. The geographic colluded-versus-competitive comparison, the one that adds finer fixed effects to the whole panel, narrows further to the eight provinces that contain both a colluded and a competitive tender, and that overlap is too thin to carry the contrast, which is why Section 7.2 rests instead on matched road packages. The matched design has a different support, forty-three pairs over 301 districts, and what binds it is the number of competitive comparison roads those pairs are built from, eight under the baseline rule and six under the tighter one that the headline figures come from. The eight provinces and the eight comparison roads are different objects that happen to share a number. It is the comparison roads that are the reason the local results are read as bounds.

4.2 The reconstructed cartel universe

The corruption in Peruvian road infrastructure has two judicially documented arms, and this paper uses both. The first is the Odebrecht (Lava Jato) record of bribery behind a set of road concessions and megaprojects, which Section 2 describes. The second, which we introduce here, is the *Club de la Construcción*, a cartel that rigged the public tenders for national road works. The two are distinct mechanisms in the same ecosystem. Odebrecht bribery operated mainly through concessions awarded by ProInversión and the officials who took the payments. The Club operated through bid rigging in the *licitaciones públicas* run by the Ministry of Transport and Communications. We treat them as two records, not one database.

Indecopi adjudicated the cartel tender by tender. Resolución 080-2021/CLC sanctions thirty-two economic groups for rigging the public tenders for national road works between 2002 and 2016, and it does so process by process. From its structured exhibits we reconstruct a universe of seventy-eight road tenders together with the firms that won or covered each one, and we recover seventy-three of them in the electronic procurement records, sixty-three of which match at high confidence on the full process signature. Award identities come from the ruling and never from those records, because the latter are incomplete for large road works. Appendix B documents the exhibits and the code reconciliation, and Appendix C the validation.

Two annexes carry the variation the paper uses. Annex 2 records each bid and carries two adjudicated labels, one on the individual offer and one on the tender that offer belongs to. The tender label is constant across the offers of a tender in every one of the 206 procedures the annex covers, and the offer label is nested inside it, so an offer flagged as collusive always sits in a tender adjudicated colluded while a colluded tender may still hold a few offers not individually flagged. A tender is classified here by its tender-level label, which is the adjudicated classification behind both the price comparison and the household contrast. Annex 3 reports the authority's own estimate of the illicit benefit on each sanctioned process. The arrangement they document is a durable one, with a designated winner on each tender surrounded by coordinated cover bids and abstentions, sustained across the period and enforced when members deviated (Appendix H).

Our firm-level measure of cartel involvement is the set of processes, values, and roles the resolution attributes to each firm. For firm f we record the number of cartelized processes it won, the number in which it acted as a cover bidder, the reference value of the processes it won where the resolution reports it, and the duration of its documented participation. The measure is a judicially documented exposure derived from the antitrust resolution and not a share of total procurement activity. It is predetermined with respect to the 2020-2021 sanction but it is not exogenous to firm size, capacity, or political connection, so we read it as exposure intensity, not randomly assigned treatment.

The reconstruction shows how the cartel divided its work. Ingenieros Civiles y Contratistas Generales (ICCGSA) carries the largest fine at 77,265 UIT, followed by Obrainsa at 72,796 UIT. The most frequent winners are JJC, ICCGSA, Construcción y Administración and Johesa, with Energoprojekt and E. Reyna tied behind them, but the winning itself is not concentrated. Twenty-eight of the thirty-two sanctioned groups hold at least one winning position, the top five hold 0.44 of them and the winner Herfindahl is 0.06, the division-by-turns pattern Section 6.2 sets out. Cover bids are documented only for the three firms whose bid exhibits are structured enough to read, Cosapi, Obrainsa and JJC, and among those three Obrainsa wins few of the processes it appears in and covers in most. We therefore treat cover bidding as documented for those firms rather than measured across the cartel.

The household analysis uses the district counterpart of this measure, the proximity-weighted exposure of each district to the geocoded tenders, whose construction is summarized in Table 3.

Table 3: Construction of the Construction Club cartel-exposure measure.

Element	Specification
Spatial unit	District (centroid), national household survey panel
Cartel tenders geocoded	76 of the 78 reconstructed road tenders
Proximity kernel	$\exp(-d_{ij}/\,10$ km) summed over nearby tenders
Distance bands (robustness)	5 km, 10 km (baseline), 20 km
Alternative intensity	count of tenders within the band
Value weighting (robustness)	reference value of nearby tenders
Treatment timing	first nearby tender year, post from that year +2
Districts ever exposed (10 km)	400 (332 cartel-only, 68 both branches)
Standardization	per one standard deviation of exposure

Notes. The exposure measure is built entirely from the judicial record, not from a procurement share, for the reasons given in Section 4.2. Two of the seventy-eight tenders could not be geocoded to a road segment and are dropped from the spatial measure. The decay, band, count, and value variants are reported in Appendix E. *Source.* Resolución 080-2021/CLC and the MTC national road network.

4.3 Mapping local exposure

To take the two records to households we measure, for each district, its exposure to each branch of the corruption, and the two measures cover different geographies (Figure 3).

For the concession branch we use the accessibility improvement each district received from the Lava Jato corridors, a network market-access calculation that compares travel times on the actual road network with a counterfactual in which the concession-corridor segments carry their pre-upgrade condition (Section 5), together with log corridor kilometers and a travel-time measure. This exposure is concentrated on the ten built concession corridors of panel A.

For the cartel branch we build a district measure from the seventy-eight reconstructed tenders, geocoding each process from its endpoint towns to district centroids and reaching seventy-six of the seventy-eight across twenty-one departments (Appendix D). For each district we compute a proximity-weighted exposure and date its treatment by the earliest nearby process. The counts depend on the distance band. Across all districts, 276 fall within five kilometers of a cartel process, 437 within ten, and 809 within twenty. Our analytic exposure uses the ten-kilometer band, under which 400 districts across twenty-three regions are exposed in the household panel. Panel B maps the wider twenty-kilometer footprint for visibility, and Appendix Figure 20 traces it growing from 258 districts by 2008 to 809 by 2016. Either way the exposure is far broader than the ten concession corridors and most of it lies away from the Lava Jato corridors, the independent geographic variation that lets the two branches tell distinct household stories in Section 7.

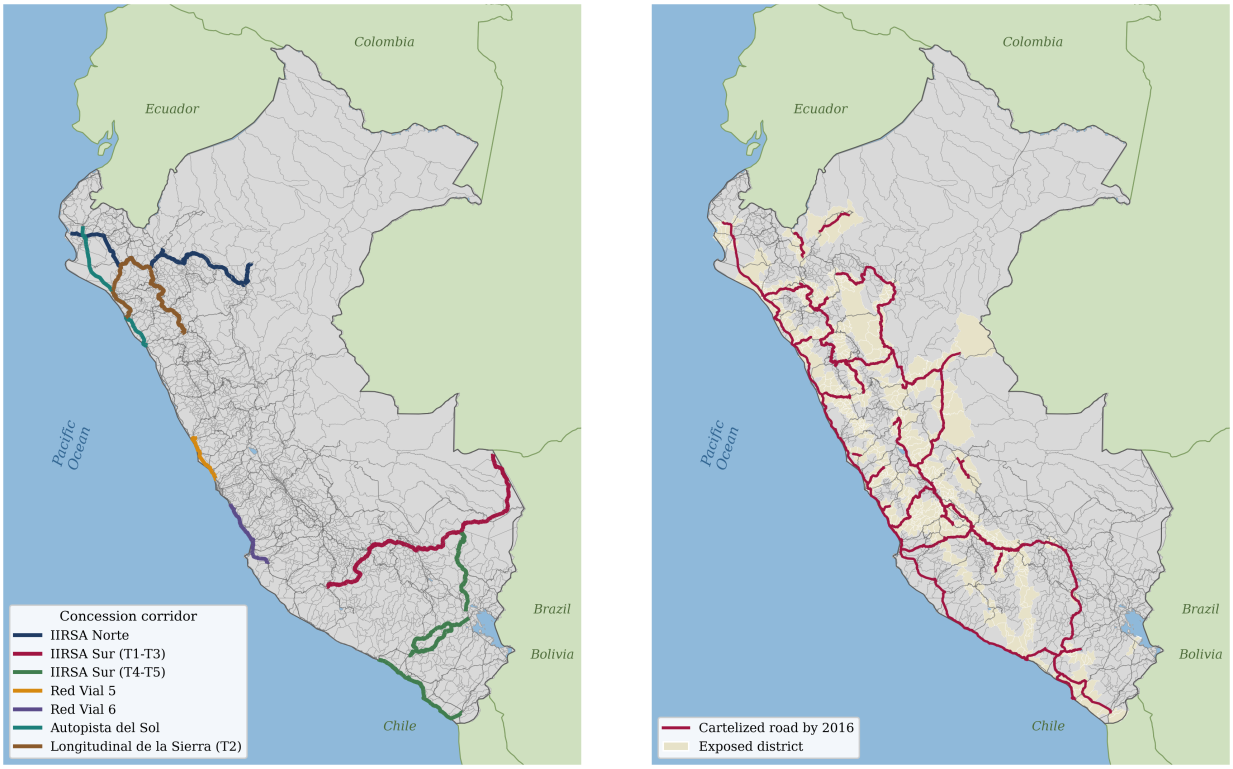

Figure 3: Geography of the two records. Panel A: Lava Jato concession corridors. Panel B: Construction Club cartel exposure by district.

Notes. Panel A shows the built concession corridors over the national road network. Panel B shades districts within twenty kilometers of a reconstructed cartel process, with the cartelized routes snapped to the actual road network overlaid and distinct from the national network in grey. Neighboring countries in green, the Pacific in blue. *Source.* Resolución 080-2021/CLC, INEI district boundaries, MTC road network, OSITRAN concession records.

5 Empirical strategy

The paper has five pieces. For the twenty-five Odebrecht projects we compare cost overruns by documented-bribe status,

$$\text{Overrun}_j = \alpha + \beta\,\text{Bribe}_j + X_j'\gamma + \varepsilon_j,$$

as means, medians, and a cross-section with controls, assessed with exact and small-sample inference.

We measure the accessibility each district received from corridor completion as the change in market potential,

$$\Delta \ln MA_d = \ln MA_d^{\text{post}} - \ln MA_d^{\text{pre}}, \qquad MA_d = \sum_m \text{Mass}_m\, T_{dm}^{-\theta},$$

where T_{dm} is travel time on the road network and the pre network sets the concession-corridor segments to their pre-upgrade condition (Section 4). A placebo on random comparable corridors validates that the gradient is specific to the actual corridors, so this is a validation of the measure, not a first stage. We relate accessibility to household outcomes in the ENAHO panel,

$$y_{dt} = \beta\,(\Delta\,\widetilde{\ln M}A_d \times \text{Post}_{dt}) + \gamma\,(\Delta\,\widetilde{\ln M}A_d \times \text{Constr}_{dt}) + \alpha_d + \lambda_{r(d)t} + \varepsilon_{dt},$$

with district effects α_d, region-by-year effects $\lambda_{r(d)t}$, Post keyed to each corridor's opening, Constr covering the years between the start of works and that opening, and corridor-level standard errors. The construction window is entered so that the post-opening coefficient is not read off years when the works were still underway, and γ is estimated but not reported. The regressor is scaled to one standard deviation of the accessibility gain among exposed districts, $\Delta \widetilde{\ln MA_d} = \Delta \ln MA_d / \mathrm{sd}(\Delta \ln MA \mid \Delta \ln MA > 0)$, so β is the response to a one-standard-deviation gain and is comparable across the four exposure measures of Table 8. The fixed effects do not solve the endogenous placement of roads, and we treat the result as associational.

The cartel exposure enters the same panel. For district d we build a proximity-weighted exposure $\mathrm{Exp}_d^{\mathrm{Club}}$ to the seventy-six geocoded cartel processes and a treatment date from the earliest nearby process, and estimate y_{dt} on $\mathrm{Exp}_d^{\mathrm{Club}} \times \mathrm{Post}_{dt}^{\mathrm{Club}}$ with the same fixed effects, clustering by region. A joint district panel then characterizes the broad geographic associations of the two records together,

$$\begin{aligned} y_{dt} = \alpha_d + \lambda_{r(d)t} &+ \beta_1 \, (\mathrm{Exp}_d^{\mathrm{LJ}} \times \mathrm{Post}_{dt}^{\mathrm{LJ}}) + \beta_2 \, (\mathrm{Exp}_d^{\mathrm{Club}} \times \mathrm{Post}_{dt}^{\mathrm{Club}}) \\ &+ \beta_3 \, (\mathrm{Exp}_d^{\mathrm{LJ}} \times \mathrm{Post}_{dt}^{\mathrm{LJ}})(\mathrm{Exp}_d^{\mathrm{Club}} \times \mathrm{Post}_{dt}^{\mathrm{Club}}) + \varepsilon_{dt}, \end{aligned}$$

where β_1 is the Lava Jato corridor effect, β_2 the Construction Club cartel effect, and β_3 the incremental effect of joint exposure over and above the two separate ones. Both exposures are continuous, so a district's total is $\beta_1 \mathrm{Exp}^{\mathrm{LJ}} + \beta_2 \mathrm{Exp}^{\mathrm{Club}} + \beta_3 \mathrm{Exp}^{\mathrm{LJ}} \mathrm{Exp}^{\mathrm{Club}}$, which equals $\beta_1 + \beta_2 + \beta_3$ at one standard deviation of each and scales with the doses otherwise. What it never equals is β_3 alone. Because the cartel onset is staggered across years, we estimate the dynamics with four estimators, two-way fixed effects, Callaway–Sant'Anna (Callaway and Sant'Anna, 2021), de Chaisemartin–D'Haultfœuille (de Chaisemartin and D'Haultfœuille, 2020), and Borusyak, Jaravel and Spiess (Borusyak et al., 2024), and rely on the two staggered-robust estimators for the identifying reading. The cartel's geography is not randomly assigned, so we read β_2 and β_3 as strong descriptive associations.

Each coefficient is a distinct estimand. The overrun β is a conditional mean difference across twenty-five projects, not an average treatment effect. The accessibility and cartel coefficients are dose-response slopes, the change in a district's outcome per standard deviation of accessibility gain and of cartel exposure, identified off the timing and geography of corridor openings and of nearby cartel tenders. Two treatment clocks run in the panel. The corridor branch separates a construction window, when works are underway, from the opening date when traffic begins, and keys $\mathrm{Post}^{\mathrm{LJ}}$ to the opening. The cartel branch keys $\mathrm{Post}^{\mathrm{Club}}$ to the first year a cartel tender appears near the district, with the post period opening two years after that year to allow for the construction lag, which is the timing Table 3 reports and the one used throughout. A district can sit on both clocks, and β_3 reads how far the joint response departs from the sum of the two separate ones.

The corruption-specific household question needs a comparison that holds the road fixed, and the competition authority's tender-level adjudication supplies one. Among road-exposed districts we separate exposure to the tenders Annex 2 records as colluded from exposure to those it records as competitive, bid by the same firms on the same class of works,

$$y_{dt} = \alpha_d + \lambda_{g(d)t} + \delta_C \, (\mathrm{Exp}_d^{\mathrm{Col}} \times \mathrm{Post}_{dt}) + \delta_K \, (\mathrm{Exp}_d^{\mathrm{Comp}} \times \mathrm{Post}_{dt}) + \varepsilon_{dt},$$

where the contrast $\delta_C - \delta_K$ is the corruption-specific component net of the road, and the geographic effect $\lambda_{g(d)t}$ runs from department-by-year to province-by-year as we tighten the control. The identifying variation is whether a district's nearby tenders were rigged or competitively bid. Because that variation is scarce, with only eight provinces containing both, the geographic comparison has the right sign but not enough common support to separate corruption from the road. The corruption-specific household design therefore rests on matched road packages, pairing each colluded tender with a comparable competitively procured one, the specification Section 7.2 carries.

6 Procurement cost and distortion

Corruption is visible first in what the State paid and in how the tenders were allocated. We take the fiscal cost of the bribed projects, then the allocation and the price the cartel extracted, and then ask what the adjudicated record implies for detecting collusion elsewhere.

6.1 The Lava Jato concession branch

The Lava Jato branch asks what the bribed projects cost and whether the corridors they built repaid the localities they crossed. The cost comparison runs on all twenty-five Peruvian Odebrecht projects, fourteen in transport, eight in water and sewerage and three in energy, which is a wider set than the road concessions the local analysis follows. Treatment is a bribe documented in a judicial ruling, taken from the Odebrecht case record assembled by Campos et al. (2021). We code three categories instead of a clean-versus-corrupt binary, a confirmed documented bribe, judicial implication without a project-specific payment, and no documented evidence. “No documented evidence” is not the same as clean, since the investigations reveal projects selectively, and we keep the categories separate.

Of the twenty-five Odebrecht projects in Peru that Campos et al. (2021) code, the ten with a confirmed documented bribe ran cost overruns averaging 53 percent of initial investment, against 17 percent for the fifteen projects not documented as bribed, with medians as wide, 47 against 8. That comparison pools the two non-bribe categories, and separating them gives 29 percent for the five judicially implicated projects and 11 percent for the ten with no documented evidence, so the ordering across the three categories is monotone (Table 20).

Because a pooled comparison group hides which of the two non-bribe categories does the work, we estimate the overrun on two category dummies with the ten no-evidence projects as the omitted baseline (Table 4). Against that explicit baseline the documented-bribe projects run +42 percentage points higher without controls, +26 once a transport indicator, log initial investment and year enter, and +28 when sector fixed effects replace the transport indicator. The five judicially implicated projects sit between the two, +18 points above the baseline without controls, but that middle category is estimated off five projects and is never distinguishable from zero, and neither is the difference between the implicated and the bribed ($p = 0.23$ without controls). The binary specification in Table 19 reports the same fiscal fact against a mixed comparison group that pools the implicated with the no-evidence projects, +36 points unconditionally and +22 with controls, which is why its gap is smaller than the +42 measured against the no-evidence projects alone. With twenty-five projects all of this describes the difference in fiscal cost and is not an estimated causal elasticity.

Table 4: Cost overruns by documentation category (descriptive, $N = 25$).

	(1) no controls	(2) +transport, size, year	(3) +sector FE, size, year
Judicially implicated	18.0	13.5	12.6
	(16.9)	(20.2)	(18.6)
Documented bribe	42.1***	25.9**	28.4**
	(12.3)	(11.0)	(13.1)
Transport		21.7	
		(15.1)	
log initial investment		11.4**	9.1
		(4.8)	(5.7)
Year		-4.3*	-4.3*
		(2.1)	(2.3)
N projects	25	25	25
R-squared	0.320	0.501	0.514
Bribe minus implicated (pp)	24.0	12.4	15.8
p, bribe = implicated	0.228	0.592	0.440

Notes. Cross-sectional regression of the cost overrun (percentage points of initial investment) on two documentation-category dummies, with the ten projects carrying no documented evidence as the omitted baseline, $N = 25$ Odebrecht projects in Peru. Column (2) uses the control set of column (2) of Table 19 and column (3) replaces the transport indicator with a full set of sector fixed effects. That control set is the one behind the conditional gap in Table 20, but the comparison group differs, since this table holds the judicially implicated projects in a separate category while the robustness table contrasts the ten documented-bribe projects with the other fifteen pooled, which is why the coefficients are +28.4 here and +25 there. The last two rows report the difference between the documented-bribe and judicially-implicated categories. Heteroskedasticity-robust standard errors in parentheses. Under HC3 the documented-bribe coefficient has $p = 0.003$ without controls, $p = 0.11$ in column (2) and $p = 0.16$ in column (3). ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$. *Source.* Odebrecht–Peru case data, Campos et al. (2021).

The unconditional gap holds up across the small-sample inference twenty-five projects allow, randomization inference over the bribe labels at $p = 0.005$, a Webb-weight wild bootstrap at $p = 0.005$, a rank-sum test at $p = 0.002$, and a leave-one-megaproject-out range of $[25, 40]$, collected in Table 20. The conditional gap is the exception, and under an HC3 standard error it is no longer distinguishable from zero at conventional levels ($p = 0.12$), so the robustness claim covers the raw difference and not every specification. Documentation is plausibly endogenous to the overrun, since a project that ran far over budget draws more scrutiny and is more likely to surface a documented bribe, which would inflate the gap. The direction of this bias is not clean. The December 2016 cross-country Odebrecht plea agreement that opened the Peruvian record was driven by the United States proceedings and not scrutiny of any single Peruvian project's cost, which limits selection at the level of which cases surfaced, but it does not discipline which Peruvian project drew a documented bribe, and treating the no-evidence projects as clean cuts the other way. We therefore read the gap as a descriptive magnitude, not a signed bound, and measure excess investment relative to initial budgets, not a rent. The bribed projects were also built by consortia that include firms sanctioned in the procurement cartel (Section 2), so the gap measures the cost of the oligopoly's documented-corruption projects and not the effect of bribery net of collusion.

6.2 Cartel allocation and the competitive discount

The Construction Club records show the allocation mechanism and its price. From the seventy-eight reconstructed tenders (Section 4) we record, for each firm, the processes it won, the processes in which it acted as a cover bidder, and the reference value of the works it won. We measure contract allocation and capture, not firm profits. Table 5 reports the firm-level counts.

Table 5: Cartel allocation and contract capture (top twelve firms by processes won).

Firm	Won proc.	Cover proc.	No-supp. / unc.	Ref. value won (S/. M)	Departments	Indep. evid.
JJC	14	28	8	1,111	Amazonas, Ancash, Apurimac +17	27
Iccgsa	11	0	0	1,102	Amazonas, Apurimac, Ayacucho +6	7
CASA	7	0	0	757	Arequipa, Cajamarca, Cusco +6	4
Johesa	7	0	0	713	Ayacucho, Cajamarca, Huancavelica +3	2
E. Reyna	5	0	0	553	Arequipa, Cajamarca, Lambayeque +4	1
Energoprojekt	5	0	0	816	Cajamarca, Cusco, Huancavelica +3	3
Obrainsa	4	38	0	–	Amazonas, Ancash, Apurimac +16	18
Constructora Malaga	4	0	0	933	Apurimac, Ayacucho, Lima +3	3
Queiroz Galvao	4	0	0	559	Cajamarca, Cusco, Huanuco +2	3
Altesa	3	0	0	23	Ancash, Cajamarca, Cusco +1	2
Aramsa	3	0	0	406	Huanuco, Pasco, Ucayali	1
Camargo Correa	3	0	0	374	Cajamarca, Huanuco, Lambayeque +1	2
Reconstructed processes						78
Share of wins, top 5 firms						0.44
Firms with documented cover bids						3
Herfindahl index of wins						0.06
Reference value of cartelized tenders (S/. bn)						6.3

Notes. Top twelve firms by number of processes won, with summary statistics over all thirty-two sanctioned groups. Ranking on wins rather than on total documented involvement is why Cosapi, with two wins but twenty-five documented cover-bid processes, does not appear here and does appear among the most involved firms in Figure 4. Cover bids are documented only for the three firms with structured apoyo exhibits. The full firm table is in Table 24. Won and cover columns count processes in which the firm was a member of the winning consortium or a documented cover bidder. Reference value won is a lower bound given partial value coverage. *Source.* Resolución 080-2021/CLC.

The structure is one of role division more than win concentration. A win here is membership in the winning consortium, which is how the annex records the award, and thirty-one of the fifty-nine tenders with a recorded winner went to a consortium of two to four firms. Counted that way wins are fairly diffuse, with a Herfindahl index of 0.06 and the top five groups taking 0.44 of documented winning positions, so no single firm dominates the awards, though part of that spread is firms appearing on the same ticket and not only taking turns. That is the internal organisation Asker (2010) documents for a bidding cartel, a designated winner supported by the others and the work shared out over time rather than captured by one firm. What the cartel organizes is the assignment of roles, a designated winner for each tender surrounded by coordinated cover bids. JJC is in the winning consortium of fourteen of the fifty-nine tenders whose winner the annex records and Ingenieros Civiles y Contratistas Generales of eleven, and the thirty-seven tenders with a reported reference value sum to about 6.3 billion soles. The cover-bid counts carry a caveat, since they are documented in full only for the three firms with structured bid exhibits, Cosapi, Obrainsa, and JJC, so the apparent concentration of cover bids in those firms partly reflects which records we could read. We therefore read the cover-bidding as coordinated support around designated winners (Figure 4), not as a full ranking of which firms covered most.

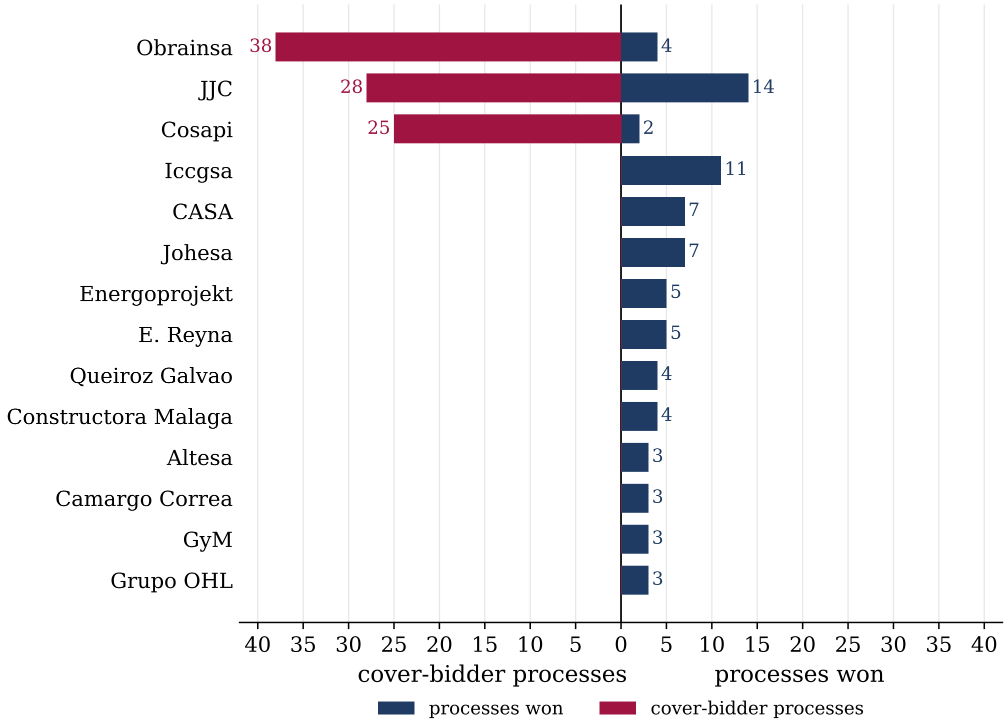

Figure 4: Role concentration: cover-bidder and won processes by firm.
Notes. Processes won (right) and processes in which the firm was a documented cover bidder (left), top firms by total documented involvement. A co-participation network of the firms is in Appendix Figure 18. *Source.* Resolución 080-2021/CLC.

The same records show what the coordination cost. Annex 3 of the resolution reports the competition authority's own estimate of the illicit benefit, and these sum to about 780 million soles. The annex lists one row per sanctioned firm per process, 193 rows over 121 processes, and 54 of those processes carry more than one sanctioned firm, so the reference value has to be counted once per process and not once per row. On that basis the sanctioned caseload carries about 16 billion soles of reference value and a median estimated overcharge of 7.1 percent, against 4.8 percent for the ratio of the two totals. The illicit benefit itself is genuinely firm-level and adds up across rows, so the 780 million is unaffected.

That figure carries a caveat. It is not a tender-level estimate. Forty-four of the sixty-eight colluded tenders that carry a positive benefit are assigned exactly 7.885 percent of the reference value, and most of the rest are simple fractions of the same rate, so the series is an administrative parameter and not an independent measurement of harm tender by tender. The 7.1 percent median is close to that administrative rate for the same reason. We use the annex for the aggregate and read nothing into its variation across tenders. The bid record makes the mechanism visible. Across the 145 of the 206 procedures in Annex 2 whose bids can be used, the winning bid on a colluded tender is a median 108 percent of the reference value, while the winning bid on a competitive tender by the same firms is a median 90 percent, a ten-point discount to the State. Both medians are unchanged on the deduplicated sample that Section 6.3 describes. Peruvian procurement law rejects a bid for works below 90 percent or above 110 percent of the published reference value, so the cartel's signal is the suppression of the competitive discount, pushing the winning bid to the top of that band, where almost all colluded winners sit while competitive winners go to the floor. The band binds in the record. Of the 145 winning bids, 143 sit inside it and two do not, one at 87.0 and one at 121.2 percent, which the annex does not explain and which we leave as recorded. The cartel both assigned the winners and erased the competitive discount, the designated-winner-and-cover-bid pattern that Porter and Zona (1993) and Conley and Decarolis (2016) recover statistically from bid distributions and that the resolution here establishes tender by tender. Colluded tenders also drew fewer bidders, a median of three against four. Figure 6 shows the two distributions barely overlapping.

The gap is not a composition effect. Table 6 regresses the winning bid share on a colluded indicator. The raw difference is 10.8 points. Year effects, terrain, works type and the log reference value leave 8.7 points, and adding the number of bidders leaves 7.5. Holding the winning firm fixed leaves 8.1 points, and holding the firm and the year fixed together leaves 4.1, every column significant at one percent. Fourteen firms won both colluded and competitive tenders, and those fourteen identify the last two columns. Figure 5 shows them one at a time. The same contractor that discounts to the floor of the band on a competitive tender bids near the ceiling when the tender it wins was rigged, by an average of eight points. The price gap is therefore a property of the tender and not of which firms turned up to bid on it.

Tenders repeat within winning firms, so those last two columns are re-inferred at the firm level. The estimation sample there holds the nineteen firms that won more than one tender, all of which define clusters, while the fourteen that won on both sides are the ones that identify the coefficient. Clustering on the winning firm barely moves the standard errors, from 0.021 to 0.022 with firm effects alone and from 0.016 to 0.018 with firm and year effects, and a Webb-weight wild cluster bootstrap over the nineteen returns $p = 0.004$ and $p = 0.042$. Two-way clustering on firm and year does not yield a usable variance matrix at this sample size and is not reported.

Table 6: Winning bid relative to the reference value, colluded versus competitive tenders.

	(1)	(2)	(3)	(4)	(5)	(6)
Colluded tender	0.1078***	0.0735***	0.0868***	0.0750***	0.0810***	0.0411***
	(0.0161)	(0.0149)	(0.0138)	(0.0139)	(0.0210)	(0.0158)
Year		Yes	Yes	Yes		Yes
Terrain			Yes	Yes		
Works type			Yes	Yes		
Log reference value			Yes	Yes		
Bidders				Yes		
Winner firm					Yes	Yes
Tenders	122	122	122	122	94	94
R^2	0.355	0.637	0.683	0.718	0.425	0.727

Notes. The outcome is the winning bid divided by the published reference value, one observation per tender, from Annex 2 of Resolución 080-2021/CLC. Bids outside 40 to 160 percent of the reference value are dropped as transcription errors, a filter on the recorded ratio and not the admissible band, which is 90 to 110 percent for works and which all but two of the estimation sample respect. The 145 tenders of Section 6.3 are the annex codes carrying a flagged winning bid. This sample is built on a canonical procurement code that collapses optical-character variants of the same code, requires a positive reference value, and takes the lowest recorded offer as the winning bid where the annex flags no winner. That leaves the 122 distinct tenders estimated here, 99 colluded and 23 competitive. It is not a subset of the 145. It shares 117 tenders with them and adds 5 for which no winner is flagged, and dropping those raises the colluded coefficient from 0.108 to 0.116 in column (1) and from 0.041 to 0.054 in column (6), so keeping them is the conservative choice. The headline screen sample of Section 6.3 is the 145 itself. Its robustness version, which additionally requires a well-formed transport-ministry code, holds 118. Neither is nested in the sample estimated here, since a screen needs the buyer identified and a price comparison does not. Columns (5) and (6) restrict to the 19 firms that won more than one tender and absorb winner-firm effects. All of them enter the estimation and define the clusters used for inference, while the colluded coefficient is identified by the 14 of them that won both a colluded and a competitive tender, since a firm on one side only contributes nothing to a within-firm contrast. Heteroskedasticity-robust standard errors in parentheses. Clustering on the winning firm gives 0.0216 and 0.0177 in columns (5) and (6), and a Webb-weight wild cluster bootstrap over the 19 firms gives $p = 0.004$ and $p = 0.042$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

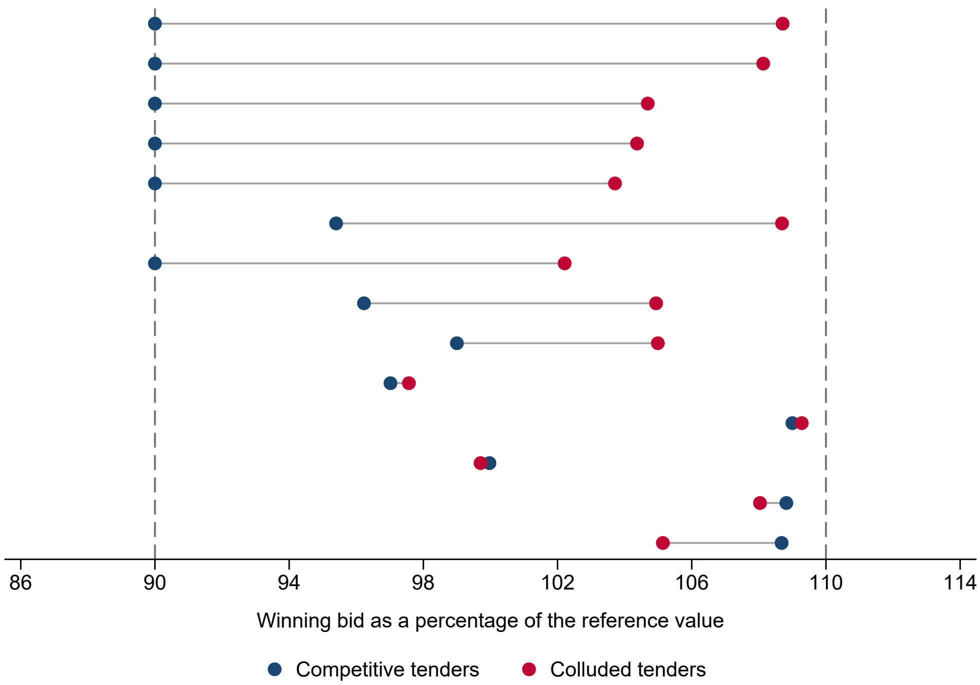

Figure 5: Winning bid within firm, colluded versus competitive tenders.
Notes. Each row is one of the fourteen firms that won both a colluded and a competitive tender, with its average winning bid on each type, expressed as a percent of the reference value. Rows are sorted by the within-firm gap. Dashed lines mark the floor and ceiling of the admissible bidding band for works, 90 and 110 percent of the reference value. *Source.* Anexo 2 of Resolución 080-2021/CLC.

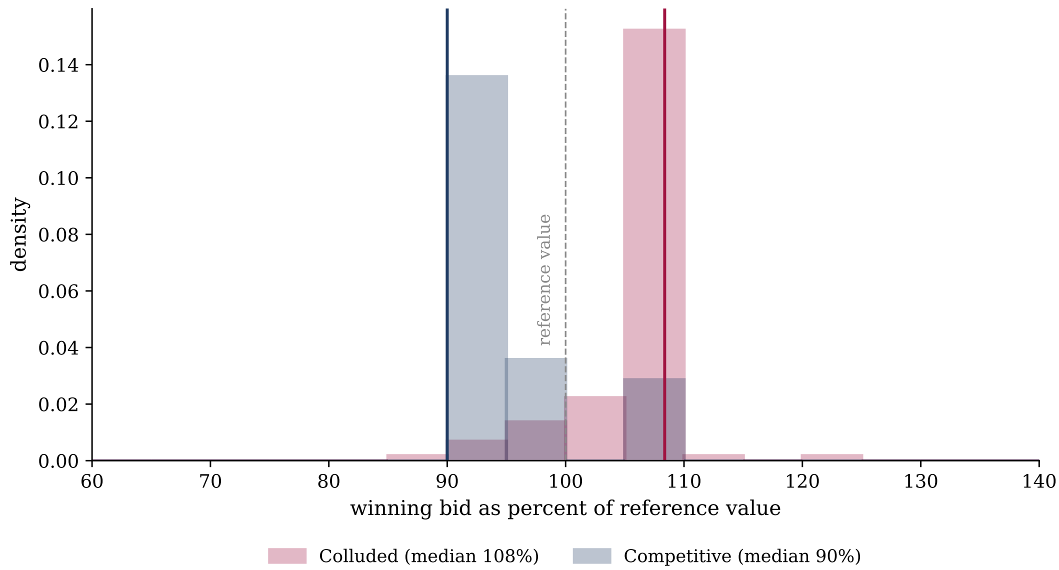

Figure 6: Winning bid as a percent of the reference value, colluded versus competitive tenders. *Notes.* Distribution of the winning bid, one per tender, as a percent of the reference value, for the 117 colluded and 28 competitive tenders with usable bids, out of the 206 procedures Annex 2 adjudicates, bid by the same cartel firms. A winning consortium lists each member as a row, so we collapse to the tender and take the offer once. Vertical lines mark the medians, 108 and 90 percent. The dashed line is the reference value. Bids are trimmed to the 40 to 160 percent range to drop transcription outliers. That range is a filter on the recorded ratio and not the admissible band, which is 90 to 110 percent and which 143 of these 145 winning bids respect. *Source.* Anexo 2 of Resolución 080-2021/CLC.

The scheme is visible tender by tender. Figure 7 plots the bids in example tenders. In the colluded ones the winner sits at or above the reference value with cover bids stacked just around it, the designated-winner pattern the resolution documents. In the competitive ones the winner gives a discount and the bids spread below.

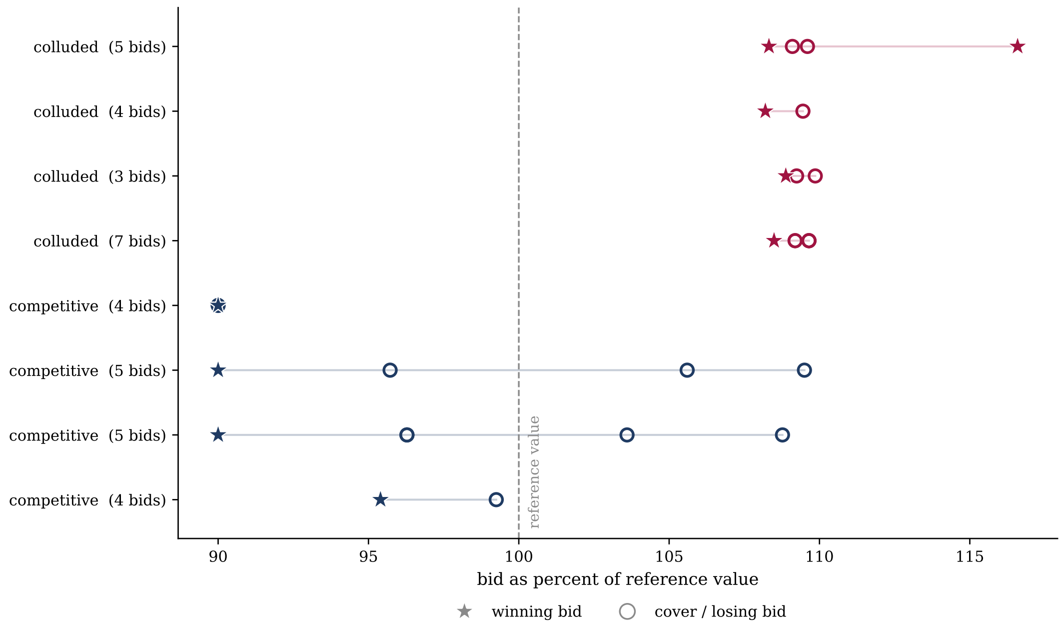

Figure 7: The cover-bid scheme in example tenders, from the bids recorded in Anexo 2. *Notes.* Each row is one tender, each marker a bid as a percent of the reference value, with the winning bid starred and the cover or losing bids hollow. Colluded tenders (top) show the winner at the ceiling with cover bids around it, competitive tenders (bottom) show a winning discount and a wider spread. Every winner shown sits inside the 90 to 110 percent band. One losing bid in the top panel is recorded above it, one of the twenty such cases among the 571 bids the annex records. *Source.* Anexo 2 of Resolución 080-2021/CLC.

Because Indecopi adjudicated each tender, the level gap can be checked against the verdict. Section 6.3 takes the detection literature's screens to these bids and scores them against that ground truth.

6.3 Detecting collusion against the adjudicated record

The collusion-detection literature builds statistical screens to flag the designated-winner-and-cover-bid structure from bid data (Porter and Zona, 1993; Conley and Decarolis, 2016; Kawai and Nakabayashi, 2022; Chassang et al., 2022). Fazekas et al. (2026) test a battery of them against a large sample of known cartels, and enforcement cases are the usual benchmark. Here the verdict is tender by tender, we hold the bids of the same tenders the authority adjudicated, and that set is the one whose price and surrounding districts the rest of the paper measures. A screen can therefore be scored on the same objects the corruption was established on.

We take three screens to the Annex 2 bids. The first is a level screen, the winning bid as a share of the published reference value, which asks whether the competitive discount was suppressed. The second is the dispersion screen in common use, the coefficient of variation of the bids, which asks whether the bids were coordinated tightly. The third is a thin-competition screen, the number of bidders. Instead of a single cutoff, we treat each as a continuous classifier and trace its full receiver operating characteristic against Indecopi's classification across the 145 tenders that a screen can be computed on, 117 colluded and 28 competitive. These are the classified tenders whose bids carry a plausible bid-to-reference ratio and an identified winning bid. The unit is the tender code as the annex prints it, and the optical character recognition behind the annex leaves 27 of the 145 malformed, some of them corrupted variants of a code that also appears intact, some truncated fragments, and six naming a regional or municipal buyer instead of the transport ministry. Collapsing to well-formed ministry codes leaves 118 tenders, 93 colluded and 25 competitive. The level screen loses about a hundredth of its area under the curve and the dispersion screen is unchanged. The duplication almost doubles the count of single-bid tenders, from 21 to 40, the one channel by which it could have manufactured the dispersion result. That is a different filter from the one the geography uses, which starts from the 119 road works whose location can be audited at all and keeps the 63 that resolve to a usable corridor (Appendix F), because a tender can be scoreable without being locatable and locatable without carrying a usable winning bid.

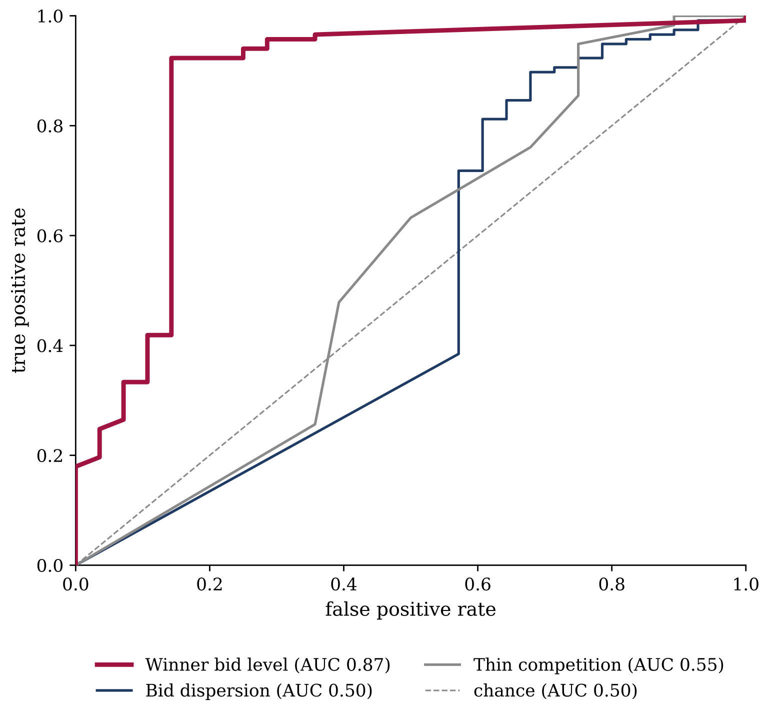

Figure 8: Receiver operating characteristic of three collusion screens against the adjudicated label. *Notes.* Empirical ROC over the 145 classified tenders, 117 colluded and 28 competitive, so each curve is a staircase and the ties from single-bid tenders and integer bidder counts appear as its flat and steep segments. Each curve traces the true-positive against the false-positive rate as the screen's threshold varies, scoring tenders as colluded, and the diagonal is chance. The winner-bid level bows to the top-left (AUC 0.87), bid dispersion tracks the diagonal (AUC 0.50, no better than chance once single-bid tenders carry their zero dispersion), and thin competition is barely above it (AUC 0.55). *Source.* Anexo 2 of Resolución 080-2021/CLC.

The level screen dominates. Its area under the curve is 0.87, against 0.50 for the dispersion screen, no better than chance once the single-bid tenders that carry zero dispersion are included, and 0.55 for thin competition. Restricted to the multi-bid tenders where dispersion is naturally measured the dispersion screen reaches only 0.62, still far below the level screen, so its weakness does not come from that treatment. Bootstrapping over tenders, the level screen's AUC exceeds the dispersion screen's by a gap whose 95 percent interval is $[0.23, 0.51]$, and its margin over the thin-competition screen is $[0.21, 0.45]$. An analytic test for the difference between correlated areas under the curve (DeLong et al., 1988) agrees, rejecting equality of the level and dispersion screens at $z = 5.3$ ($p < 0.001$) and of the level and thin-competition screens at $z = 5.3$ ($p < 0.001$). Table 7 reads off illustrative operating points at simple thresholds, where the level screen classifies 88 percent of tenders correctly while the dispersion screen at its median cut does no better than a coin flip.

Table 7: Collusion screens validated against the adjudicated label.

Screen	Accuracy	False positive	False negative
Winner bids above the reference value	0.88	0.14	0.12
Low bid dispersion (CV below median)	0.47	0.21	0.61
Few bidders (three or fewer)	0.61	0.50	0.37

Notes. Each screen predicts a tender as colluded, and the prediction is compared to Indecopi's adjudicated classification across 145 tenders with at least one classified bid (117 colluded, 28 competitive). Accuracy is the share correctly classified, the false-positive rate is among competitive tenders, the false-negative rate among colluded tenders. *Source.* Anexo 2 of Resolución 080-2021/CLC.

The ranking generalizes beyond this case. Chassang and Ortner (2019) study collusion when bids are constrained, which is the environment a published reference value creates. Where bids are anchored that way, the cartel's gain is to erase the discount the State would otherwise capture, so the signal lives in the level of the winning bid and not in the spread of the losing ones that a dispersion screen is built to detect. A screen tuned to variance can miss a cartel that coordinates on price instead of rank. The adjudication makes this checkable against the assignment that Chassang et al. (2022) and Kawai et al. (2023) must infer.

7 Local returns to corrupt procurement

Whether the localities these roads were built to serve received a commensurate return is a separate question, and the two branches answer it differently.

7.1 The concession corridors

The bribed corridors delivered no detectable local dividend. In the ENAHO district panel, estimated against log corridor kilometers, two market-access measures, and a travel-time measure, none of the twelve post-opening estimates on household income, consumption, or poverty is distinguishable from zero, and the nulls are stable to dropping the gold-mining departments, dropping Lima, guarding against spillovers, and leaving out each corridor. For the conservative market-access measure a ten-percent gain in access moves household income by about -1.5 percent and consumption by -1.8 percent, both indistinguishable from zero, with a corridor-level wild-cluster bootstrap returning an income p of 0.55 and a consumption p of 0.46. The corridors measurably improved accessibility, and no local welfare gain from it is detectable here. The design is associational (Section 5), so this is a failure to find a dividend and not a demonstration that none exists. Table 8 reports the twelve estimates.

Table 8: Concession corridors and household outcomes after opening.

	Log income	Log consumption	Poverty rate
Log corridor kilometres	-0.0041	-0.0087	+0.0058
	(0.0112)	(0.0096)	(0.0060)
Market access, conservative	-0.0161	-0.0187	+0.0148
	(0.0149)	(0.0133)	(0.0135)
Market access, mid	-0.0054	-0.0088	+0.0103
	(0.0171)	(0.0154)	(0.0129)
Travel time to the provincial capital	+0.0075	-0.0012	-0.0121
	(0.0216)	(0.0161)	(0.0109)

Notes. Each cell is the post-opening coefficient on one exposure measure for one outcome, from a separate regression of the ENAHO district panel with district and region-by-year fixed effects and standard errors clustered on the corridor, reported in parentheses. Each measure is scaled to one standard deviation among exposed districts, so the cells are comparable across measures. That is why the conservative market-access row reads -0.0161 while the text quotes -1.5 percent: a standard deviation of that measure is 0.105 log points, an access gain of about eleven percent, and the text states the effect of a ten percent gain. The p-values the text reports for that row are the corridor-level wild bootstrap, which at roughly ten corridors is weaker than the analytic t implied by the standard errors here. The construction-period coefficient is entered alongside and not shown. The four exposure measures are log corridor kilometres, two market-access measures built from the road network, and travel time to the provincial capital. None of the twelve is distinguishable from zero, the largest t statistic in the table being 1.41. Section 5 sets out why this design is read as associational. *Source.* ENAHO *sumaria* district panel matched to the concession corridors.

A null on average could hide redistribution toward the better-connected places a corridor links, the pattern Faber (2014) finds for China's highways, where activity concentrates in the central cities a road joins. We test it two ways. Splitting the corridor-exposed districts by whether they are provincial-capital nodes or peripheral leaves no extra gain at the nodes, with a capital interaction of $+0.003$ on income ($p = 0.84$), so the null is not a local loss masked by an upstream gain captured at the capitals. The districts the corridor physically crosses do fare worse than the adjacent catchment on income, by a relative -0.035, consistent with an interregional trunk road that passes through the rural periphery without developing it, though with ten corridors this contrast is imprecisely identified and consumption does not echo it. The corridors show no detectable local dividend and at most a faint pass-through penalty, not a redistribution toward connected centres.

One mechanism behind a costlier project that delivers no more is renegotiation after award. Across the seven audited concessions the three with a documented bribe average eight distinct contract addenda against five for the rest, consistent with bribed concessions being reopened more often, in the spirit of Decarolis (2014) and Coviello et al. (2018) on renegotiation absorbing value. With seven concessions this is descriptive, not a test. On the cartel side the analogous exercise is not available, because the electronic procurement records carry no addenda data for the large road licitaciones, the same gap that blocks a firm-level outcome design.

7.2 The Construction Club cartel branch

The overcharge is the price of the corruption on the tenders the cartel controlled. Whether it also reached the households around those tenders is the harder question, and we turn to it now. The broad cartel-procurement exposure is associated with adverse outcomes. Table 21 places the corridor and cartel measures in the same district panel with district and region-by-year fixed effects and region-clustered standard errors. The corridor measure is null throughout, while the cartel measure is negative per one standard deviation of exposure, -0.016 on income and -0.017 on consumption, and survives controlling for the corridor measure. Read alone this looks like a cartel effect, but nothing in the comparison yet separates the corruption from the road.

The cartel measure compares districts near a cartelized tender to districts near no such tender, which includes districts near no major road tender at all. Building the identical proximity exposure for the Ministry of Transport national road licitaciones of the same period that are *not* in the cartel set produces the same association, -0.016 on income and -0.018 on consumption, indistinguishable from the cartel coefficient. The adverse association at this resolution is a feature of being near a large road tender, competitively or corruptly procured, and the broad measure recovers the road effect.

The competition authority's tender-level adjudication allows a sharper comparison. Annex 2 adjudicates each tender the cartel firms entered as colluded or competitive, the same firms bidding on the same class of works, so comparing district exposure to colluded tenders against exposure to competitive ones holds the firms and the road class fixed and varies only whether the tender was rigged. Table 22 enters both exposures together. With district and department-by-year fixed effects and province-clustered standard errors, colluded exposure is adverse on all three outcomes, -0.027 on income, -0.024 on consumption, and $+0.013$ on poverty, each significant under a province wild-cluster bootstrap. Competitive exposure is not distinguishable from zero on any outcome at the department and province levels, the one exception being consumption inside the eight both-arm provinces, where it is positive and significant ($+0.019$, $p = 0.02$). Unlike the broad dose the contrast survives a control for the size of the nearby roads, where the colluded coefficients hold or strengthen.

The corruption-specific reading turns on whether the colluded coefficient differs from the competitive one, the quantity $\delta^C - \delta^K$ defined in Section 5. Table 22 now reports that difference as an estimated row. It is adverse for income and consumption on every rung of the ladder, between -0.02 and -0.07, and it is not distinguishable from zero at the department level, with $p = 0.14$ on income, 0.32 on consumption, and 0.60 on poverty. It approaches conventional levels only for consumption, inside the both-arm provinces ($p = 0.07$) and once road size is controlled ($p = 0.09$), and both of those cells move away from significance under a province wild-cluster bootstrap ($p = 0.13$ and $p = 0.14$). The first of the two is restricted to the eight provinces holding both kinds of tender, where the bootstrap is the only defensible inference at that cluster count, while the second keeps the department-by-year sample. The colluded arm carries the adverse association, but the point estimate keeps its size as the geographic fixed effects tighten while the interval widens, so this design does not separate the two arms. Appendix E bounds the contrast and shows the broad-exposure event study by four estimators.

A closer counterfactual pairs each colluded tender with a comparable competitive one. For every colluded road package we take the competitive road tender closest in geography, award year, reference value, and terrain, treat the districts within ten kilometers of the colluded tender as exposed and those within ten kilometers of its competitive match as the control, and stack the pairs so that a pair-by-year fixed effect compares the two arms within the same matched package and year (Appendix F). This replaces the cross-region competitive control with a matched road of the same kind. Over a four-year window around the tender, exposure to a colluded package is associated with household income about fourteen percent lower and consumption about eleven percent lower than its competitive match (Table 23).

How much of this is real turns on the level at which the design randomizes, and here the design is thinner than the district count suggests. The forty-three matched pairs draw on only eight distinct competitive comparison roads, and those roads are reused unevenly, one of them anchoring twelve pairs. Clustering by district treats each reuse as fresh information. Table 23 therefore reports the same coefficients under a ladder of inference levels. Clustering by matched pair leaves all three sharply significant, but clustering on the eight comparison roads, and the wild-cluster bootstrap appropriate to eight clusters, leaves only income significant and then only at the ten percent level ($p = 0.06$), with consumption at $p = 0.18$ and poverty at $p = 0.18$. A matched-pair permutation test that flips the colluded and competitive labels within each matched pair puts income at $p = 0.03$, consumption at $p = 0.06$, and poverty at $p = 0.07$. Roads were not assigned to collusion by lottery, and the permutation test is valid only under the assumption that, conditional on the match, which arm of a pair was rigged is exchangeable under the null. That assumption is weaker than a design-based one, but it is the assumption the matching is meant to make plausible, and the test is informative about whether the estimate could arise from the label pattern alone. Under the baseline match, then, income is the estimate that holds at every level and consumption and poverty do not clear that bar.

The baseline matches are loose. Median separation between a colluded road and its comparison is 143.5 kilometers, the median gap in log reference value is 0.80, and only 57 percent of pairs share a terrain category, because the matcher allows a 400 kilometer caliper and penalizes terrain instead of requiring it. We therefore rebuilt the design once under materially tighter calipers, 200 kilometers, four years and a hard bound of 0.80 on the log reference-value gap, holding the match score, the exposure band, the event window, the two-year lag and the estimator fixed so that only the choice of comparison road changes. Requiring identical terrain is not feasible here. Only ten competitive road tenders survive geocoding, their terrain mix does not span the colluded ones, and a 150 kilometer same-terrain rule leaves a single pair. The tighter rule retains eighteen pairs on six comparison roads and cuts the median reference-value gap from 0.80 to 0.28.

The estimates grow under the tighter match. Income moves from -0.144 to -0.168, consumption from -0.114 to -0.160, and poverty from $+0.068$ to $+0.109$, and all three are significant at five percent under the matched-pair permutation test. Under the six-road wild bootstrap poverty clears five percent and consumption and income clear ten (Table 9). Appendix F.4 shows that this holds across the whole ladder of calipers and not only at the rule we report, with the same sign on every rung and the reported rule sitting mid-range.

Tightening the calipers also shrinks the sample, from forty-three pairs on eight comparison roads to eighteen on six, so a larger estimate is consistent both with better counterfactuals and with the surviving subset having larger effects. The ladder establishes the direction. The estimates keep their sign on every rung and are larger at the tight end than at the loose one, with one non-monotone rung in the middle. We read the household incidence as adverse and substantive. The tighter sample rests on eighteen pairs and six comparison roads, few enough that the choice of bootstrap weights matters: Rademacher weights admit only $2^6 = 64$ sign vectors and cannot resolve p-values below about 0.02, so we use Webb weights throughout (Webb 2023), which is the standard remedy below roughly a dozen clusters.

Table 9: Matched road-package estimates under the baseline and a tighter matching rule.

	Baseline match			Tighter match		
	Cons.	Income	Poverty	Cons.	Income	Poverty
Colluded × Post	−0.114	−0.144	+0.068	−0.160	−0.168	+0.109
p-value by level of inference						
District clusters	0.019	0.006	0.042	0.002	0.014	0.001
Matched pairs	0.001	0.000	0.002	0.001	0.005	0.001
Comparison roads	0.111	0.026	0.128	0.030	0.028	0.023
Wild bootstrap, comp. roads	0.179	0.061	0.180	0.075	0.099	0.041
Matched-pair permutation	0.062	0.028	0.067	0.003	0.017	0.014
Matched pairs		43			18	
Competitive comparison roads		8			6	
Observations		4,257			1,506	
Median separation (km)		143.5			132.1	
Median $\lvert\Delta \ln$ ref. value$\rvert$		0.80			0.28	

Notes. The same stacked difference-in-differences, district-within-pair and pair-by-year fixed effects, ten kilometer exposure band, four-year window and two-year post lag, under two matching rules. The baseline matches each colluded road to the competitive road minimising a score in distance, award year, log reference value and a terrain penalty, inside a 400 kilometer caliper. The tighter rule keeps that score and imposes hard calipers of 200 kilometers, four years and 0.80 in log reference value, so only the admissible comparison set changes. Requiring the same terrain category is infeasible, as only ten competitive road tenders survive geocoding. The wild bootstrap clusters on the competitive comparison roads with Webb weights and 9,999 replications. Webb rather than Rademacher because with six or eight clusters Rademacher admits only 2^G sign vectors, a resolution of 1/64 and 1/256, too coarse to separate the outcomes at conventional levels. The matched-pair permutation test flips the colluded and competitive labels within each pair over 1,999 draws and compares studentized statistics. *Source.* ENAHO *sumaria* matched to Anexo 2.

The dynamics confirm the timing. Under comparison-road inference, which is the level this design supports, no pre-treatment lead is individually distinguishable from zero for income or consumption, and under the baseline rule a joint test of the three does not reject either, at $p = 0.35$ for income and $p = 0.48$ for consumption. Under the tighter rule income and poverty pass the same test while consumption rejects at $p = 0.031$, the one pre-period imbalance the design carries, which is why the tighter consumption path is read more cautiously than the income one it is quoted beside. Clustered on the matched pair the consumption lead at $k = -4$ does reach conventional levels, at -0.106 with $p = 0.040$, and Appendix F shows that this single lead is what the pair-level joint test picks up. And the adverse path is already open in the tender year and deepens through year four, with poverty rising in step (Figure 9). Poverty carries one elevated pre-period lead at year minus four, so we read its level more cautiously than the income and consumption paths. The dynamics survive inference at the level of the comparison roads. Re-estimating the same event study with a wild-cluster bootstrap over the eight comparison roads leaves every pre-period coefficient far from significance, income at p of 0.66, 0.60 and 0.42, while the post-period path stays adverse and strengthens, from $p = 0.09$ at the tender year to 0.03 by year four. We read the design as identifying a moderate adverse incidence and not a precise point estimate, and Appendix F audits the geocoding and shows the result holds out leaving any single comparison road.

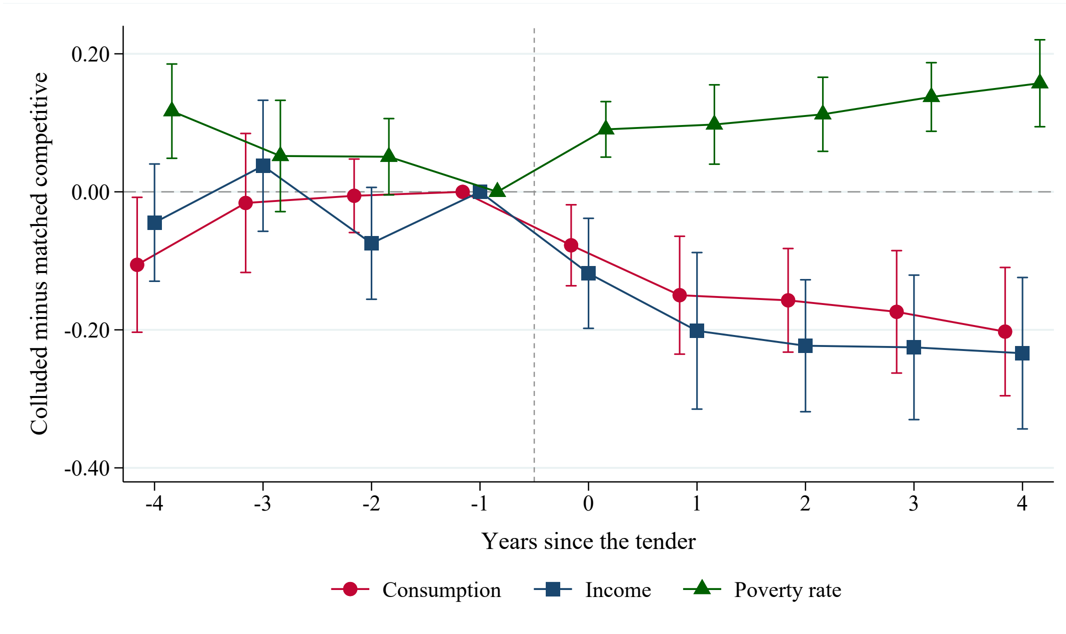

Figure 9: Matched road-package event study, colluded minus matched competitive exposure.
Notes. Within-pair event study of the matched road-package design for consumption, income, and poverty, per district, with district-within-pair and pair-by-year fixed effects and district-clustered 95 percent intervals, on a four-year window around the tender. Income and consumption are in log points, poverty in rate points. The reference is the year before the tender (dashed). The figure plots every event year in the window, while the post period of the matched estimates begins at year two, which is the construction lag the design allows and which Section 8 separates from the early years zero and one. Income and consumption fall and poverty rises after the tender, with one elevated poverty lead at year minus four. *Source.* ENAHO *sumaria* matched to Anexo 2.

How far does that contrast extend? The design defines a road's catchment as the districts within ten kilometres, and the estimate compares those districts with the ones around the matched competitive road. It does not say what happens to the wider region. To ask that we widen the catchment to forty kilometres, keep the same pairs, tender dates, window, lag and inference, and collapse every district inside it into one area observation per road-year, weighting by population so that a district of forty surveyed households does not count the same as one of four. The area effect is -0.008 on income and -0.004 on consumption under that weight, neither distinguishable from zero. Every cell stays small and none becomes distinguishable from zero under the other two weights, surveyed households and surveyed persons, or when the area outcome is built from logs rather than from levels and then logged.

The immediate-radius result therefore says where the burden falls, and the wider area gives no evidence of a loss: the area estimates are small, negative and far from significance, so a regional effect of the size the ten-kilometre contrast would imply can be ruled out, while a smaller one cannot. The natural next question is whether activity relocated, and the ring-by-ring profile in Appendix F is suggestive of that, adverse against the route and positive twenty to forty kilometres out. We do not rest anything on it. Only twenty of the forty-three pairs have both arms populated in every ring, and on that fixed set the near-road income coefficient falls from -0.111 to -0.029, so a material part of the apparent profile is which pairs enter which ring rather than distance. What the evidence supports is the local contrast and the absence of a regional counterpart of the same size.

Where, then, does the cost of the corruption land? It is identified on the procurement side, in the larger overruns on the bribed projects and the cartel's overcharge on the tenders it controlled, while its incidence on the localities is identified less sharply, by the matched road-package design and from few comparison roads. One natural alternative does not show up. Displacement of other local public investment by exposure to the colluded tenders is not detectable, at the district level or at the region level, where the estimate is negative and imprecise. That is a failure to detect crowd-out and not a demonstration that none occurred, so it weakens a local displacement reading without establishing that the whole fiscal cost lands on the national budget (Appendix K).

8 Labour-market incidence and timing

The matched design says households near a colluded road are poorer than households near a comparable competitively procured one. It does not say through what. The household aggregates are consistent with fewer people working and with the same people earning less, and the two have different implications, so where the loss lands comes first.

We take the matched design unchanged, the same pairs, the same fixed effects, the same window and lag, and replace the outcome. From the employment and income module of the same household survey we build district-year measures of the employed share of the working-age population, weekly hours among the employed, and labour earnings, split into earnings from dependent employment and earnings from independent work. Table 10 reports them under both matching rules, since nothing below should rest on one of them.

Table 10: Where the household loss appears: employment and earnings margins.

	Tighter match		Baseline match	
Outcome	Estimate	p	Estimate	p
Household aggregates				
Log household income	−0.168	0.005 [0.028] {0.099}	−0.144	0.000 [0.026] {0.061}
Log household consumption	−0.160	0.001 [0.030] {0.075}	−0.114	0.001 [0.111] {0.179}
Employment margin				
Employed share, working age	−0.012	0.364 [0.236] {0.075}	−0.026	0.003 [0.035] {0.049}
Weekly hours, employed	+1.19	0.105 [0.414] {0.784}	−1.47	0.001 [0.121] {0.153}
Self-employed share	+0.022	0.138 [0.350] {0.279}	+0.017	0.086 [0.353] {0.341}
Earnings margin				
Log labour income per adult	−0.197	0.010 [0.057] {0.043}	−0.173	0.000 [0.038] {0.043}
Log labour income, employed	−0.183	0.013 [0.058] {0.049}	−0.139	0.002 [0.104] {0.074}
Log dependent earnings	−0.200	0.064 [0.215] {0.025}	−0.096	0.215 [0.477] {0.474}
Log independent earnings	−0.194	0.016 [0.016] {0.058}	−0.221	0.000 [0.007] {0.041}
Matched pairs	18		43	
Competitive comparison roads	6		8	

Notes. The matched road-package design of Section 7.2 with the outcome replaced. District-within-pair and pair-by-year fixed effects, ten kilometer exposure band, four-year window, two-year post lag. Three p-values are reported for each estimate: analytic and clustered by matched pair first, then analytic and clustered on the competitive comparison roads in brackets, then a Webb-weight wild cluster bootstrap over those same roads in braces, on eight clusters under the baseline rule and six under the tighter one. The bootstrap is the inference the household estimates of Section 7.2 rely on at this cluster count, and it is the one we read. The employed share is of the working-age population, hours are weekly hours among the employed, and earnings are annual labour income, reported per working-age adult and, separately, among the employed. Dependent earnings come from wage employment and independent earnings from own-account work and employers. Money amounts are nominal, which the pair-by-year fixed effects absorb. The household aggregates are repeated from Table 9 for comparison. *Source.* ENAHO *sumaria* and the employment and income module, matched to Anexo 2.

Throughout this section and the last, a coefficient on a logged outcome is read as a percentage of the same size, which overstates the proportional change by a few percent of itself at these magnitudes.

The loss is concentrated in earnings. Labour income per working-age adult falls by about twenty percent under the tighter match and seventeen under the baseline, and labour income among those who are working falls by about eighteen and fourteen. That the estimate barely moves when the sample is restricted to people who are employed matters, because it means the shortfall is not an artifact of counting zeros for people who lost work.

The employment margin is present but poorly determined. The employed share is adverse under both rules and clears conventional levels under the comparison-road bootstrap in both, at $p = 0.075$ and $p = 0.049$, so we do not read it as a null. Its magnitude more than doubles between the rules, from -0.012 to -0.026, and weekly hours are positive under the tighter rule and negative under the baseline with neither distinguishable from zero, so the size of any employment response is not identified here. The design supports the earnings margin as the larger of the two and the better determined.

Within earnings, the most stable component is independent work. Earnings from independent activity fall by about nineteen percent under the tighter match and twenty-two under the baseline, and it is the only component of the earnings decomposition that clears conventional levels under clustering on the competitive comparison roads in both specifications, as the household income aggregate also does. Dependent-employment earnings move in the same direction but are weaker. They survive neither analytic comparison-road clustering under either rule nor the bootstrap under the baseline, and clear conventional levels only under the tighter rule and only in the bootstrap. The two magnitudes are also close. Labour income per working-age adult falls by 0.197 and household income per capita by 0.168. The denominators differ, so the two are not components of an identity and this is a comparison of size and not a decomposition of one into the other.

The earnings result does not deliver an industry. Splitting independent earnings into agriculture, manufacturing, commerce, transport and other services gives a decline concentrated in commerce under the tighter match and in transport and manufacturing under the baseline, so the two rules disagree about which sector carries it. The one feature stable across both is that agriculture does not fall, its coefficient being positive and insignificant either way. The loss sits in non-agricultural own-account activity, and the data do not identify which part of it.

This locates the incidence without identifying a mechanism. The household income loss does not operate primarily through employment destruction. It appears in labour earnings among those who remain employed, with the most robust decline concentrated in independent earnings.

Where the loss lands is one question and when it opens is another, and the second discriminates between readings that the first cannot. If corruptly procured roads deliver a worse road, the shortfall should appear once the works are finished and the road is carrying traffic. If instead the procurement and implementation of the works disturbs the local economy, it should appear earlier. The baseline post period begins two years after the tender, precisely to allow for the construction lag, so it corresponds to $k = 2$ onwards. Here we widen the estimation window to reach back before it, entering the early years $k = 0$ and 1 as a separate block alongside the baseline window $k = 2$ to 4, with the pre-period as the omitted baseline. Event time runs from the tender year, not from the physical start of works, which is unobserved, so the early window is defined by the procurement clock. Table 11 reports both blocks and a test of their equality.

Table 11: When the loss appears: early post-tender window against expected-delivery window.

	Early post-tender $k = 0, 1$	Expected-delivery $k = 2, 3, 4$	Equality p
Tighter match			
Log household income	−0.135	−0.233	0.016 {0.051}
	(0.088) [0.082] {0.264}	(0.011) [0.037] {0.132}	
Log household consumption	−0.128	−0.221	0.016 {0.029}
	(0.049) [0.099] {0.336}	(0.002) [0.029] {0.115}	
Poverty rate	+0.073	+0.144	0.006 {0.030}
	(0.027) [0.028] {0.043}	(0.001) [0.024] {0.044}	
Log labour income per adult	−0.138	−0.264	0.013 {0.031}
	(0.195) [0.164] {0.125}	(0.027) [0.085] {0.054}	
Log independent earnings	−0.322	−0.350	0.632 {0.509}
	(0.007) [0.005] {0.008}	(0.005) [0.008] {0.025}	
Baseline match			
Log household income	−0.136	−0.203	0.007 {0.040}
	(0.002) [0.110] {0.076}	(0.000) [0.047] {0.064}	
Log household consumption	−0.090	−0.153	0.004 {0.089}
	(0.032) [0.220] {0.229}	(0.003) [0.143] {0.195}	
Poverty rate	+0.048	+0.089	0.014 {0.169}
	(0.071) [0.308] {0.299}	(0.004) [0.172] {0.193}	
Log labour income per adult	−0.186	−0.253	0.005 {0.035}
	(0.000) [0.077] {0.061}	(0.000) [0.052] {0.044}	
Log independent earnings	−0.252	−0.329	0.060 {0.126}
	(0.000) [0.033] {0.024}	(0.000) [0.010] {0.033}	

Notes. The matched design with the post period split into $k = 0, 1$ and $k = 2, 3, 4$, both measured against the omitted pre-period, entered in one regression. Analytic p-values clustered by matched pair in parentheses, analytic and clustered on the competitive comparison roads in brackets, and a Webb-weight wild cluster bootstrap over those roads in braces. The last column is a Wald test that the two block coefficients are equal, reported analytically and, in braces, under the same bootstrap. *Source.* ENAHO *sumaria* and the employment and income module, matched to Anexo 2.

The loss is already there in the early window. Every outcome is adverse in the first block under both matching rules, so the pattern is not one of an effect that waits for the road to open. For the household aggregates the second block is larger, and significantly so, with equality rejected at p between 0.004 and 0.016, so the shortfall deepens instead of reversing once the works should have been delivered. Independent earnings behave differently and more sharply. Under the tighter rule they fall by about thirty-two percent in the early window and thirty-five afterwards, magnitudes that are statistically indistinguishable ($p = 0.63$). Under the baseline rule the two blocks are -0.252 and -0.329 and equality is rejected at $p = 0.060$, so the flat profile belongs to the tighter rule and not to both. Independent earnings reach most of their decline in the early window on either reading, and under the tighter rule they reach essentially all of it.

Small-cluster inference sharpens this. Under a wild cluster bootstrap over the comparison roads, the early coefficient on independent earnings is significant under both matching rules, at $p = 0.008$ and $p = 0.024$, and it is the only outcome whose early block clears conventional levels under both. The early coefficient on household income does not under the tighter rule ($p = 0.26$) and just does under the baseline ($p = 0.076$). The deepening between the two blocks survives the bootstrap for income and consumption under the tighter rule and for income under the baseline, with equality rejected at p between 0.029 and 0.051, while the poverty deepening survives only under the tighter rule. What opens early, on the inference the design can support, is the independent-earnings margin specifically.

The timing does not fit a channel that runs only through the finished road, since such a channel would leave the early window quiet and it is not quiet. It is consistent with disruption to local economic activity that begins around procurement and implementation and does not reverse. In much of rural Peru the independent workers are the small traders, transporters, farmers and microenterprises who stand to gain most from a road that improves market access, so their earnings are a natural place for a shortfall to appear first. Detours and closures, worse temporary access to markets, uncertainty around a contested work, lost trade around the site, and a scarring reading in which small businesses lose customers and do not fully recover are all consistent with this shape, and the data cannot separate them.

Two readings compete with this one and can be examined. The first is that colluded roads are simply worse roads, so the localities they serve get less. The second is that the district averages move because the people in them change, through population loss or sorting, and not because incumbents earn less. Appendix Table 15 takes each to the data.

For road condition we use the national road network as recorded at the end of 2014, assigning to each tender the network within two kilometres of its geocoded route and restricting the comparison to tenders awarded between 2004 and 2010, which leaves at least four years between award and the reading. Delivery dates are not observed for this universe, as the end of this section explains, so that window is chosen to make a post-delivery reading likely and does not establish one. There is no systematic adverse pattern. The share of road in the worst condition class is if anything lower on the colluded side, the not-good share likewise, and the paved share is indistinguishable. These estimates rest on four independent competitive comparison roads, and a condition recorded in 2014 also reflects later maintenance, use and rehabilitation, so it does not measure quality as delivered. The evidence shows no persistently worse physical condition on the colluded roads.

For population and composition we compare the same matched pairs across the 2007 and 2017 censuses. Population growth, the share of residents who lived elsewhere five years earlier, and the age, education and employment composition are all indistinguishable between the two arms, and the 2007 migration difference is null as well, providing no indication of differential pre-existing mobility. One limit deserves stating plainly. The census question records where current residents lived five years earlier, so it identifies arrivals and not departures, and a flat net population is consistent with offsetting inflows and outflows. We therefore do not claim that the affected households stayed. The result does make a purely compositional account of the household estimates less likely, since such an account would ordinarily leave a visible mark on population or composition and none is present. Both sets of estimates, and the geocoded crosswalk behind the road comparison, are in Appendix F.5.

The delivery timeline itself cannot be tested. Asking whether colluded works were built later or executed worse requires execution records at the level of the work. The electronic procurement system holds contract records for only a handful of the annex road tenders that carry an identifier, and the recorded amendment counts carry no variation. The national investment-closure file has the right fields, completion status, physical start and end dates, approved against updated cost, but not the right universe. Of its twenty-four thousand transport investments only twenty-one are classified as national roads, the file being overwhelmingly local and neighbourhood works, and none of the corridors at issue here appear in it. The audit office registry that does track physical progress is not reachable from outside the country. The physical-delivery channel is therefore undetermined by the available administrative data, not tested and dismissed.

9 Conclusions

Two legal records let us study Peruvian road corruption with the corrupt projects named in the rulings themselves. The State paid more for them. Odebrecht projects with a documented bribe went over budget by 53 percent on average, the others by 17. On the contracts the Construction Club controlled, the winner charged 108 percent of the cost estimate the government had published, while contracts that were competitively bid were won at 90. The same firm's bid was about four points higher against that estimate on a contract it had agreed in advance to win, so the difference is a feature of the contract and not of which firms happened to bid.

Pairing each cartel road with a comparable competitively bid one, the districts around the cartel road have household income about seventeen percent lower, consumption sixteen percent lower, and poverty eleven points higher. Those figures come from the stricter pairing, which rests on six competitive roads out of the ten that can be located, and that small number limits how precisely any of this can be measured.

Those figures are about the districts next to the works, not about the region. Taking the whole area within forty kilometres as one unit, we find no difference we can measure, however the districts are weighted. A regional loss as large as the local one can be ruled out, a smaller one cannot. Where the loss falls, it is mainly in earnings, and mostly among the self-employed. Fewer people are working as well, by less and with less precision. It begins within two years of the contract being awarded, which is not what one would expect if the problem were that a corrupt road is a worse road.

Two competing explanations can be checked, and neither shows up. The cartel roads are not in visibly worse condition when the network is surveyed years later, and there is no sign that the people living around them changed, either through population loss or through who moved in. What no available record covers is how the works were actually executed, so a delivery problem is untested, not ruled out.

Authorities and researchers use statistical tests to flag contracts that may have been fixed. Such tests have been checked against known cartels before, though usually case by case. Here the verdict names the contracts one at a time and we hold their bids. A test based on how much the winner charged ranks a fixed contract above a competitively bid one 87 percent of the time, while a test in common use, based on how far apart the competing bids are, does no better than a coin toss. When bids must fall within a band around a published cost estimate, a cartel does not need to bid high or to bunch its bids together. It only needs to stop discounting. The evidence of that sits in the price the winner charged, and a test that looks at the spread of the bids will miss it.

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A Reconstruction of the two judicial datasets

The paper draws on two separate judicial records, and its data contribution is uneven across them. The Construction Club side is assembled here for the first time, from the antitrust resolution's structured exhibits into a universe of tenders, firm roles and adjudicated labels. The Odebrecht side is not: the project-level fiscal data come from the case dataset of Campos et al. (2021), which already codes bribes, bribe amounts, overruns, sectors and investment, and our contribution there is to reproduce their comparison on the Peruvian subset and to geocode the corridors. This appendix documents both, so the cartel universe can be reproduced from the public ruling and the imported module can be traced to its source.

A.1 The Construction Club cartel

The source is Resolución 080-2021/CLC of the Peruvian competition authority, a document of about 2,630 pages that sanctions thirty-two economic groups for rigging the procurement of national road works between November 2002 and December 2016. The ruling does not present a single allocation table. The cartel's conduct is recorded instead in firm-specific support exhibits, each listing the tenders in which one firm submitted cover bids in favor of the designated winner. We transcribed the three structured exhibits page by page from the rendered pages, not the noisy text layer, the Cosapi exhibit (twenty-five processes), the Obrainsa exhibit (thirty-eight), and the JJC exhibit (fifty-two entries over forty-nine distinct processes). Each row records the process code, the project, the bidding consortium, the winner, and whether support was given.

We normalized the process codes, which appear in two parallel notations in the ruling, a classic licitación form and a Ministry-of-Transport subunit form, and merged the three exhibits into a deduplicated union of seventy-eight distinct road tenders. Merges were made only on a matching code together with a compatible project name and a compatible winner, and cross-notation candidates and within-exhibit collisions were resolved by hand against the ruling's narrative. Two cases illustrate the discipline. The Obrainsa exhibit lists the Túnel Callao consortium as the winner of licitación 7-2011, which the ruling states verbatim is an error referring to a different project, so we record the winner the resolution names. Licitación 11-2010 is the one documented appeal case, where the designated winner was disqualified and the award passed on appeal to another cartel member, which we code as a within-cartel reassignment, not an outside win. Across the seventy-eight tenders only one has an outside winner, which is why a design based on tenders the cartel tried and failed to control is not available.

Two annexes supply the quantities. Annex 2, on pages 2578 to 2625, is an offer-level bid database of 571 rows carrying for each bid the process code, the bidding group, the winner, the reference and offered values, and the ruling's classification of the tender as colluded or competitive, which is the variation the household contrast in Section 7.2 uses. Annex 3 reports, for each sanctioned process and firm, the reference value, the estimated illicit benefit, and the base fine, 193 rows in total. The illicit-benefit column is the competition authority's own forensic estimate of the overcharge and is the basis for the aggregate overcharge figure in Section 6.2. We read it as the authority's estimated overcharge on the sanctioned processes and not as an observed rent accruing to the firms. We validated the reconstructed universe against the national electronic procurement records, matching about four in five of the post-2004 codes on the full process signature for existence and reference value, with the pre-2004 shortfall a documented consequence of the electronic system beginning in 2004. Winners and roles are always taken from the resolution and never from the procurement records, which have structural gaps in the award data for large works. The reconstruction yields the firm-process role table, the proximity-decayed district exposure measure, and the per-process illicit benefit used in the main text.

A.2 The Lava Jato concessions

The Lava Jato branch is built from the Odebrecht record, whose collaboration agreements and prosecutions name the road concessions, the bribes, and the officials who received them. From this record we identify the set of national road concessions and which carry a documented bribe. Project fiscal cost follows Campos et al. (2021), whose comparison of bribed and non-bribed infrastructure costs we reproduce on the Peruvian subset, reading the overrun as a magnitude on the twenty-five projects the comparison allows. We treat the 2016 disclosure of the Odebrecht agreements as the event that opens the documentary record for the concession branch. It is not the treatment date for the household panel, which is keyed to corridor openings as described next.

For the local analysis we date when each corridor's improved works opened to traffic. We mined the regulator's annual reports, about one hundred documents, for opening dates and retained, for each corridor, the first verifiable opening of improved works, not the contract or exploitation date, accepting a date only when it agreed across independent reports and rejecting dates that recurred across many concessions as report artifacts. The corridor accessibility measure follows Donaldson and Hornbeck (2016). We build a noded road network, assign population-weighted origins from a gridded population surface and explicit feeder times to the network, and compute the change in market access to provincial-capital destinations from corridor-specific documented pre-improvement speeds. We disclose one defect in the underlying network, a structural break on the central highway between Huánuco and Lima that the raw network renders as a long detour, which we flag and which largely cancels in the accessibility difference. The household panel merges this exposure onto the national household survey at the district level.

B Reconstructing the cartel from Resolución 080

The cartel universe is read directly from Resolución 080-2021/CLC-INDECOPI (Expediente 001-2020/CLC), the 2,630-page antitrust ruling, not its noisy optical-character text. The resolution documents the cartel through firm-specific evidence exhibits in which each firm's own records list the tenders it participated in and the consortium that took each award. Three of these are structured process tables, and we transcribe them at the page level, the Cosapi exhibit (CSP18, twenty-five processes), the Obrainsa exhibit (OBN10, thirty-eight processes), and the JJC exhibit (JJC01, fifty-two entries over forty-nine distinct processes). Counts of exhibit rows and counts of processes differ wherever an exhibit lists a process more than once, which is why JJC appears in fifty processes in the reconstructed roles, forty-nine of them from its own exhibit and one from another. The remaining named evidence (Johesa, GyM, San Martín and others) is narrative or email, not a process table, and we use it only to corroborate participation.

Merging the three exhibits requires care because the same tender appears under different code notations across firms, a *clásico* number in one exhibit and an MTC code in another. We match on a conservative full code signature and confirm each candidate on project name and designated winner before merging, so that distinct sub-tramos and distinct lots are never collapsed by code similarity alone. Several cases were resolved by returning to the resolution text, process LP 7-2011-MTC/20 appears in one exhibit with an erroneous consortium that the resolution states corresponds to a different project, the LPN 8-2005 process splits into two awarded tramos with different winners, and the third member of the 6-2011 consortium was corrected against the resolution. The union is seventy-eight distinct processes spanning 2002 to 2016. For each firm in each process we record a role, winning-consortium member, cover bidder, cover bidder of uncertain target, or participant without support, and these roles and the designated winners are the source for the firm-level allocation in Section 6.2. Winners and roles always come from the resolution, never from the procurement data.

C Validation against the procurement records

We validate the seventy-eight processes against Peru's open procurement data, the OCDS export of the SEACE system. An early attempt matched poorly for 2015 and 2016, which we traced to our own processing, not the data. For the post-2015 regime the process code lives in the record's `officialNumber` and the work name in its `description`, and an earlier extract had kept only the title field. Re-extracting the raw OCDS with the full fields recovers the missing processes. Of the seventy-eight, seventy-three (94 percent) are found in the enriched procurement layer and sixty-three (81 percent) match at high confidence on the licitación type, the MTC sub-unit, and the work name. The unmatched remainder is four pre-2004 processes that predate digital procurement coverage and a small set with a resolution-versus-SEACE code numbering mismatch.

Two limits of the procurement data shape how we use it. The award and supplier records are empty for the large road licitaciones, so SEACE is not a usable source of winners for the works that matter, and winner identity therefore comes from the resolution while SEACE serves only to confirm existence, code, entity, work, date, and the order of magnitude of the value. For the same reason we do not construct a cartelized-share-of-procurement measure, because the per-firm award denominator is too incomplete for the cartel firms. The exposure measure is built entirely from the resolution.

D Geocoding and district exposure

We geocode each process from the endpoint towns named in its work title, matching the place names to the centroids of the INEI district polygons and resolving capital-town aliases whose district carries a different name (Aguaytía to Padre Abad, Tingo María to Rupa-Rupa, Yauri to Espinar, and similar). Seventy-six of the seventy-eight processes are geocoded across twenty-one departments, the two exceptions being a pre-2004 process with no usable endpoint and an urban grade-separated interchange with no corridor. For each of the 1,891 INEI districts we compute the count, reference value, and cover-bidder count of cartel processes within five, ten, and twenty kilometers, a distance-decayed intensity at each scale, and a treatment year equal to the earliest nearby cartel process. For the maps we snap each process to the actual MTC road network and route between its endpoints, so the cartelized routes follow real roads instead of straight lines. The endpoint-based geocoding is an approximation, and the proximity bands and decay scales in Appendix E show the household result is not an artifact of any single bandwidth.

E Household robustness

Table 12 reports the post-treatment effect of the cartel-exposure measure on household income and consumption across the robustness battery, proximity measured by a distance-decayed intensity and by a count within five, ten, and twenty kilometers, a reference-value weighted measure, district weighting by household count, alternative clustering, the three treatment cohorts, a joint drop of the two most exposed regions, and the leave-one-region-out range. Consumption is stable throughout. Income keeps its sign and magnitude everywhere but weakens under province and two-way clustering, at 0.054 and 0.074, and it is not distinguishable from zero when Ayacucho and Apurímac are dropped together, at 0.265 against 0.081 for consumption. Those two are the most cartel-exposed regions in the panel by mean exposure, so dropping both removes most of the variation the estimate rests on, and that is why the main text reads income as the more fragile outcome. Two variants are weaker than the rest. The count within ten kilometers keeps its sign but is not significant for income. Weighting nearby tenders by reference value yields estimates close to zero for both outcomes, so the association is substantially stronger for proximity-based exposure than for value-weighted exposure, and the baseline is not driven by the monetary size of nearby contracts.

Table 12: Household robustness, cartel exposure (post-treatment, per one standard deviation).

	Income		Consumption	
Specification	β	p	β	p
Decay, 5 km	−0.015	0.074	−0.017	0.004
Decay, 10 km (baseline)	−0.016	0.027	−0.017	0.003
Decay, 20 km	−0.019	0.013	−0.016	0.005
Count within 5 km	−0.019	0.045	−0.023	0.003
Count within 10 km	−0.014	0.249	−0.018	0.064
Count within 20 km	−0.026	0.011	−0.021	0.012
Value-weighted	−0.002	0.912	−0.008	0.560
Weighted by households	−0.021	0.042	−0.021	0.002
Cluster by province	−0.016	0.054	−0.017	0.007
Cluster two-way	−0.016	0.074	–	–
Cohort 2004–2008	−0.015	0.158	−0.016	0.022
Cohort 2009–2011	−0.022	0.175	−0.019	0.082
Cohort 2012–2016	−0.028	0.250	−0.027	0.137
Drop Ayacucho and Apurímac	−0.016	0.265	−0.019	0.081
Leave-one-region-out	[−0.020, −0.014]		[−0.020, −0.014]	

Notes. Post-treatment effect per one standard deviation of the cartel-exposure measure, district and region-by-year fixed effects, standard errors clustered by region unless noted. Cohorts restrict the treated set to districts first exposed in the period shown, keeping the never-exposed as controls. The two-way consumption cell is omitted because the variance estimator is singular in that specification. *Source.* ENAHO *sumaria* district panel and Resolución 080-2021/CLC.

The staggered-design estimators also pass their native pre-trend checks. The joint placebo test of the de Chaisemartin–D'Haultfœuille estimator does not reject flat pre-trends for income ($p = 1.00$), consumption ($p = 0.58$), or poverty ($p = 0.35$), and the Borusyak pre-trend test is likewise not rejected ($p \approx 0.78$ to 0.85). The one pre-trend that fails is two-way fixed effects on the poverty rate ($p = 0.04$), the specification the staggered-design critique targets, which is a further reason the main text reads the result off the staggered-robust estimators.

E.1 The staggered estimators and the contrast

The colluded-versus-competitive contrast in Table 22 cannot be estimated with the binary staggered-robust estimators, because the competitive control group lacks common geographic support. The competitive tenders fall in a small set of coastal and urban departments, while the colluded cohorts span about twenty mostly interior departments, most of which contain no competitive tender at all. The Callaway–Sant'Anna group-time estimates are correspondingly incoherent. Cohort effects on log income range from $+0.41$ to -0.18 and reach economically implausible magnitudes at longer horizons, with raw group-time effects as large as $+0.67$, and the cohorts that read most positive are precisely those whose districts were already growing fastest before any treatment, with pre-treatment income slopes of eleven to thirteen percent a year against about five percent in the controls. The positive aggregate is a weighted average of cross-regional comparisons that violate parallel trends, not a treatment effect. The de Chaisemartin–D'Haultfœuille estimator, which rests on more local comparisons, returns a null with clean placebo tests. We therefore read the contrast off the geographic fixed-effect ladder below, which conditions on region-by-year variation and returns a stable point estimate, and we treat the binary staggered estimates as uninformative here, not evidence either way.

E.2 Bounds

The colluded coefficient of Section 7.2 is adverse and distinguishable from zero at the department level. The colluded-minus-competitive difference that the corruption-specific reading requires is adverse too but is not distinguishable from zero there, with p of 0.14 on income, 0.32 on consumption and 0.60 on poverty (Table 22). Both stay adverse and of similar size as the geographic fixed effects tighten while their intervals widen (Figure 10). The colluded roads sit disproportionately in interior departments whose incomes rose through the commodity boom, while the competitive tenders cluster in a smaller set of coastal and urban provinces, so a comparison that crosses regions can confound the corruption with secular regional divergence. Moving from department-by-year to province-by-year fixed effects leaves the colluded coefficient almost unchanged in size but grows the standard error until it is no longer significant, and restricting to the eight provinces that contain both a colluded and a competitive tender returns the same adverse signs at similar magnitude with no precision, resting on twenty-seven districts. What changes across the ladder is precision, not the point estimate.

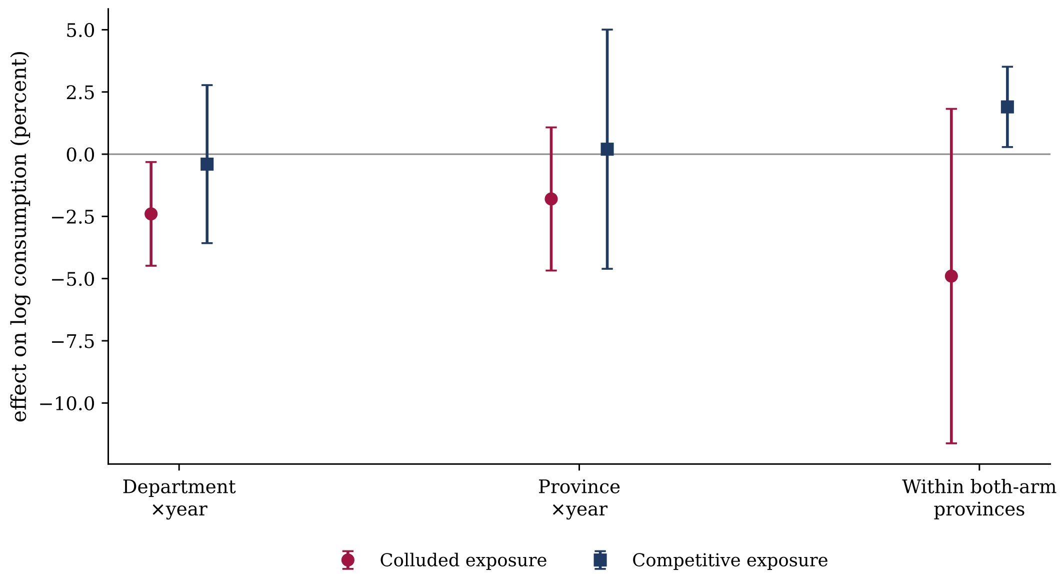
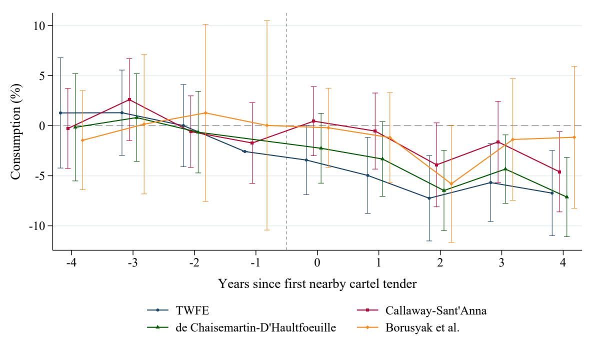


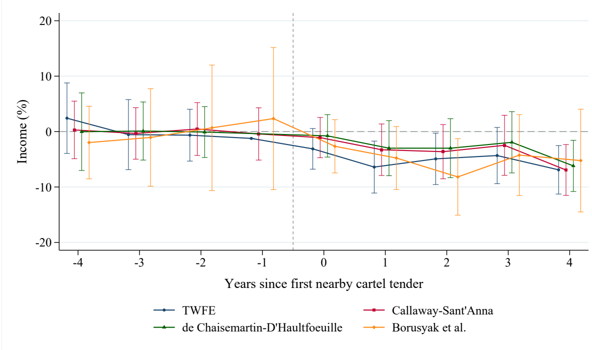
Figure 10: Colluded versus competitive exposure on consumption, across the geographic fixed-effect ladder.

Notes. Effect of colluded and competitive road exposure on log household consumption, per one standard deviation, at three geographic fixed-effect levels, with 95 percent province-clustered confidence intervals. The colluded point estimate stays adverse and of similar size while the interval widens, the precision loss from only eight provinces containing both kinds of tender. *Source.* ENAHO *sumaria* matched to Anexo 2.

Because this comparison is underpowered, we bound it. At the department level, where the colluded coefficient is significant, the minimum detectable effect at eighty percent power is about 3.4 percent for income, 2.9 percent for consumption, and 1.7 points for poverty. For selection on unobservables, the Oster (2019) statistic for the colluded coefficient, benchmarked on the competitive exposure and the road-size control, is 0.51 for income, 0.24 for consumption, and 0.48 for poverty, all below one, so unobserved selection weaker than the observed controls would suffice to move the coefficient to zero. For parallel trends, the Rambachan and Roth (2023) relative-magnitudes restriction on the broad-exposure event study (Figure 11) lets the robust confidence interval include zero once a post-treatment deviation reaches about half the largest pre-treatment violation. The broad proximity measure is the road effect, adverse but indistinguishable from competitive road exposure, and its event study by four estimators is flat in the pre-period and adverse afterward on the two staggered-robust estimators. These coarse designs bound the household incidence, and the matched road-package design of Section 7.2 identifies it more sharply.



(a) Log household consumption



(b) Log household income

Figure 11: Broad cartel-tender exposure by four estimators.

Notes. Event study of the broader road-tender exposure association on a four-year window, normalized to each estimator's pre-period mean, with district and region-by-year fixed effects and region-clustered standard errors. Under the two estimators robust to staggered timing the pre-trends are flat and the post path grows adverse. The binary staggered estimators cannot anchor the colluded-versus-competitive contrast, because the competitive control group lacks common geographic support. *Source.* ENAHO.

F The matched road-package design

This appendix documents the matched road-package stacked difference-in-differences used in Section 7.2, the geocoding it rests on, and its robustness.

F.1 Geocoding audit

The design needs the road tenders placed on the map, and the binding input is the competitive comparison set. Two Annex 2 universes appear in the paper and they answer different questions. The screen validation of Section 6.3 scores 145 tenders, those whose bids carry a plausible bid-to-reference ratio and an identified winning bid, which is what a screen needs and which does not require the work to be locatable. The geography here instead starts from the road works whose location can be audited at all, 119 tenders, 102 colluded and 17 competitive. Auditing them resolves 63 to a usable corridor, 40 in class A and 23 in class B (Table 13), and the matched design draws only on those. The 119 is the audit universe and not the number of tenders the design can place. The two counts are filters on the same annex for different purposes, not a sample restriction of one on the other, and neither is a subset of the other because a tender can be locatable without carrying a usable winning bid. We geocode every road tender in that 119-tender subset from its work title, using only pre-outcome information, the endpoint towns in the title, INEI district and province names, a small alias dictionary for capital towns whose district carries a different name, and the SEACE work descriptions. We never use household outcomes to build or tune the geocoding. Each tender is classified by the quality of its location. Class A has two or more distinct district endpoints and a clear route, class B has one district endpoint and a short corridor, and class C resolves only to a province, is an urban grade-separation interchange with no rural corridor, or has a title truncated at the source. The main design uses A and B. Table 13 reports the audit.

Table 13: Geocoding audit of the Anexo 2 road tenders.

	A (two endpoints)	B (one endpoint)	C (excluded)	Total
Colluded	32	21	49	102
Competitive	8	2	7	17
Total	40	23	56	119

Notes. Class A has two or more district endpoints, class B one district endpoint, class C only a province, an urban interchange, or a source-truncated title. The competitive C tenders are four Lima grade-separation interchanges and three with blank or truncated titles, none of which is a rural road package. The design uses the A and B tenders, ten competitive comparison roads available. *Source.* Anexo 2 of Resolución 080-2021/CLC and the SEACE procurement records.

F.2 Design and balance

For each colluded road package we take the competitive road tender closest on a standardized index of geographic distance, award year, log reference value, and sierra/selva zone, within a four-hundred-kilometer and five-year caliper. Treated districts are those whose centroid is within ten kilometers of the colluded tender, control districts are within ten kilometers of the matched competitive tender, and districts within ten kilometers of both are dropped. Stacking the pairs lets a pair-by-year fixed effect compare the two arms within the same matched package and calendar year, and a district-within-pair fixed effect absorbs the level of each district. We estimate on a four-year window around the tender, with the year before the tender as the single reference, the window chosen in advance because the household survey is noisy at the district level and the competitive comparison set is small, so distant leads and lags carry little information. The estimation keeps forty-three pairs with both arms populated, built from eight competitive comparison roads, covering three hundred one districts. Table 14 reports the match quality. The matches are a median one hundred forty-four kilometers apart and one award year apart, with a reference-value gap near a factor of two and the same terrain in three pairs of five.

Table 14: Matched-pair balance.

Match gap (colluded vs matched competitive)	Median	Mean
Geographic distance (km)	143.5	153.6
Award-year gap (years)	1	2.0
Log reference-value gap	0.80	0.86
Same sierra/selva zone (share)	–	0.57

Notes. Distribution of the gap between each colluded tender and its matched competitive tender across the forty-three estimated pairs. *Source.* Anexo 2 of Resolución 080-2021/CLC, INEI district geography.

F.3 Robustness

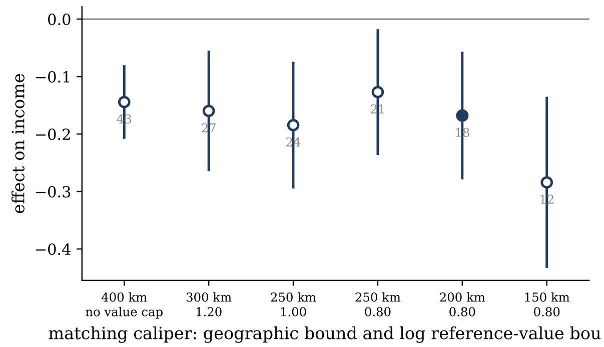
Four checks bear on whether the design is producing the result. First, leaving out any single one of the eight competitive comparison roads leaves the adverse post-period effect in place, so no one control road drives it. Second, the point estimates are stable when the window is widened from four to the full eight years after the tender, with the wider window adding the noisier late horizons and losing precision, which is why the four-year window is the reported specification. Third, the six colluded tenders that the audited geocoding adds relative to a coarser pass sit in lower-income districts than the rest, and the effect holds with or without them. Running the prior, noisier geocoding through the same code reproduces the design, so the estimation is not the source of the result.

The fourth check addresses the weight, not the standard error. Because a comparison road that anchors many pairs enters the estimate many times, and one road anchors twelve of the forty-three baseline pairs, clustering corrects the inference but leaves that road heavy in the coefficient itself. We therefore reweight each observation by the inverse of the number of pairs sharing its comparison road, so that the roads contribute equally, and separately cap reuse at three pairs per road. The estimates move modestly and keep their sign throughout. Under the reported tighter rule, income goes from −0.168 to −0.149 when the roads are weighted equally and −0.167 under the cap, consumption from −0.160 to −0.156 and −0.165, and poverty from +0.109 to +0.112 and +0.110. Under the baseline rule the shifts are larger but of the same kind, income from −0.144 to −0.131 and consumption from −0.114 to −0.096. The reused comparison road is therefore not driving the point estimate, though it does carry more of the baseline than of the tighter match.

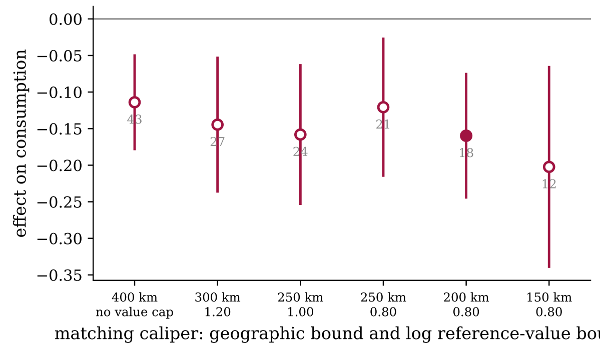
F.4 Caliper ladder

Section 7.2 reports one tighter matching rule, chosen after checking which calipers still leave a sample large enough to estimate. The full ladder is therefore shown instead of two hand-picked points. Every rung uses the same match score, geography, award year, log reference value and a terrain penalty, so the only thing that moves across rungs is which competitive roads the calipers admit, and the band, window, lag and estimator are the same throughout. The loosest rung is the paper's baseline match and the filled marker is the reported tighter rule.

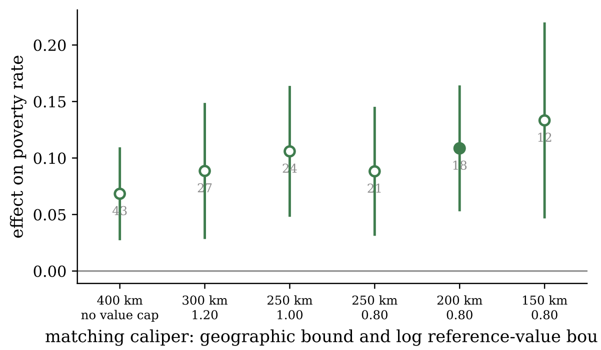
Figure 12 plots the estimate and its ninety-five percent interval at each rung, with the number of surviving pairs printed under each point. The sign is the same on all six rungs for all three outcomes, and the magnitude does not depend on landing at the reported rule, which sits in the middle of the range and not at its edge. Income runs from −0.144 at the loosest rung to −0.284 at the tightest, consumption from −0.114 to −0.202, and poverty from +0.068 to +0.133, with one non-monotone rung in the middle. Every rung is significant at five percent under pair clustering. The adverse estimates grow larger as the comparison roads are drawn from closer and from more similar reference values, and that no single caliper choice is carrying the result.



(a) Log household income



(b) Log household consumption



(c) Poverty rate

Figure 12: Matched estimates across the caliper ladder.

Notes. Each point is the post-period matched estimate at one matching rule, with the ninety-five percent interval from pair-clustered standard errors and the number of surviving matched pairs printed below. Rules run from the paper's baseline, a 400 kilometer geographic caliper with no cap on the reference-value gap, to a 150 kilometer caliper with a 0.80 cap, tightening the year caliper from five to four along the way. The filled marker is the rule reported in Section 7.2. *Source.* ENAHO *sumaria* matched to Anexo 2.

F.5 Road condition and population

Section 8 reports two alternative readings of the household result, worse delivered infrastructure and spatial sorting. This appendix gives the construction and the full estimates.

Road condition. The national road network is taken from the Via Nacional MTC layer, a snapshot dated 31 December 2014 carrying route codes, kilometre markers, surface, a condition class and geometry. It is used because the road authority's own service, the audit office public-works registry and the finance ministry investment monitor are all unreachable from outside Peru. We do not match on work names, which failed against the national investment-closure file because its universe is wrong, since only twenty-one of its twenty-four thousand transport investments are classified as national roads. Instead each Anexo 2 tender that survives endpoint geocoding is assigned the network segments lying within two kilometres of its route, and outcomes are length-weighted. Forty-eight of the sixty-three geocoded tenders match at least one segment, thirty nine colluded and nine competitive, with a median of 72 kilometres matched. Restricting to awards between 2004 and 2010 leaves thirty colluded and seven competitive tenders. Differences are taken within matched pairs and then collapsed to one observation per competitive comparison road, because the fifteen usable pairs draw on only four distinct comparison roads and a pair-level standard error would treat reuse as independent information. The paved-share difference changes sign once that correction is applied.

The layer's condition field carries five categories across its 3,467 segments, Muy Buena (69), Buena (2,415), Regular (499), Mala (380) and Sin Información (104). We recode them into the two adverse indicators the table reports. The worst-condition share counts Mala alone, 380 segments. The not-good share counts Mala together with Regular, 879 segments. Muy Buena and Buena are the good categories, and the 104 segments with no recorded condition enter neither adverse indicator, so they are pooled with the good ones, which if anything works against finding the colluded roads in worse condition. The paved share uses the layer's numeric surface field, counting its codes 1 and 2 as paved, which is 2,757 of the 3,467 segments. That field ships without a label dictionary, so the paved share is reported as a secondary outcome and read only as a null.

None of this measures quality as delivered. A condition recorded at the end of 2014 also reflects maintenance, traffic and rehabilitation in the years after the work was handed over, and delivery dates are not observed for this universe, so the exercise asks whether the colluded corridors are in observably worse shape years later and not whether they were built worse.

Population and composition. District outcomes come from the 2007 and 2017 censuses. The recent-migrant share is the fraction of residents who did not live in the district five years earlier, computed excluding the not-yet-born category. Each outcome is regressed on the colluded indicator with matched-pair fixed effects, clustered on the competitive comparison road, over the districts of the tighter match. Windows aligned to the treatment timing and the full unrestricted sample give the same answer, and the two matching rules disagree in sign on population growth, which is the signature of a coefficient at zero and not of a small effect.

Table 15: Two alternative explanations, examined.

	Estimate	Inference level
A. Road condition in 2014, colluded minus competitive		
Share of road in worst condition	−0.125	$t = -1.46$, 4 roads
Share not in good condition	−0.208	$t = -1.33$, 4 roads
Paved share	+0.078	$t = +0.42$, 4 roads
B. Population and composition, colluded minus competitive		
Log population change, 2007 to 2017	−0.035	$p = 0.38$
Recent-migrant share, 2017	−0.005	$p = 0.75$
Recent-migrant share, 2007	+0.001	$p = 0.97$
Working-age share, 2017	+0.006	$p = 0.53$
Secondary or higher share, 2017	+0.016	$p = 0.65$
Employed share, 2017	+0.037	$p = 0.22$

Notes. Panel A compares the national road network within two kilometres of each tender as recorded at 31 December 2014, for tenders awarded 2004 to 2010, differenced within matched pairs and then collapsed to one observation per competitive comparison road. A condition recorded in 2014 also reflects later maintenance, use and rehabilitation, so it is not a measurement of quality as originally delivered. Panel B regresses each census outcome on the colluded indicator with matched-pair fixed effects, clustered on the competitive comparison road. Both panels use the tighter matching rule. *Source.* Via Nacional MTC 2014, INEI censuses of 2007 and 2017.

F.6 How much the estimate depends on any one comparison road

The matched design draws on eight competitive comparison roads, and the forty-three pairs are not spread evenly across them. LP-1-2008 anchors twelve pairs, LP-15-2007 nine, LP-31-2006 and

LP-8-2007 seven each, LP-12-2016 four, LP-11-2007 two, and LP-3-2014 and LP-9-2008 one each. It is fair to ask whether the result is a property of one or two donors. Table 16 answers that in three ways.

Dropping each comparison road in turn and refitting moves income between -0.103 and -0.167 around the reported -0.144, consumption between -0.062 and -0.156 around -0.114, and poverty between 0.040 and 0.100 around 0.068. All twenty-four refits keep their sign and stay significant at five percent. No single donor carries the estimate.

Matching without replacement is the conservative version of the same question. An optimal one-to-one assignment on the same matching score gives nine pairs on nine distinct roads, so the number of pairs equals the number of independent controls. The signs survive at -0.078, -0.087 and 0.064, and none is significant, which is what nine clusters buy. The point estimates are about half the baseline and the standard errors about two and a half times as large. This is the floor the design can support.

The pre-period needs the same inference unit as the post period. Clustered on the pair, the joint test over leads -4 to -2 rejects for all three outcomes, at $p = 0.049$, 0.032 and 0.009. But the pair is not the unit the paper uses for the post estimates, and forty-three pairs built on eight roads overstate how much independent information the pre-period holds. Tested jointly at the comparison-road level with a wild cluster bootstrap on Webb weights, the same null does not reject for any outcome, at $p = 0.35$ for income, 0.48 for consumption and 0.22 for poverty. Under the tighter rule income and poverty also pass, at 0.30 and 0.40, while consumption rejects at 0.031, which is the one pre-period imbalance the design carries and the reason we read the tighter consumption path more cautiously. The analytic road-clustered test is not a useful guide here, moving from 0.014 to 0.000 across outcomes on six clusters, which is the small-cluster behaviour the bootstrap exists to correct.

Taking the leads one at a time shows where the pair-level rejection comes from. Income alternates sign across the three leads, -0.045, $+0.038$, -0.075, which is not a trend. Consumption and poverty concentrate almost everything at $k = -4$, -0.106 and $+0.117$, and decay to near zero by $k = -2$. Dropping one comparison road at a time moves the $k = -4$ consumption coefficient between -0.064 and -0.156, so it is not one donor either, and the observations at that lead fall disproportionately in 2005, 2007 and 2008. What the pair-level test picks up is one isolated lead concentrated in a few early calendar years, and not a slope that carries into the post period.

Table 16: Dependence of the matched estimate on individual comparison roads.

	Log income	Log consumption	Poverty rate
Reported matched estimate	−0.1443	−0.1140	0.0684
Leave-one-comparison-road-out range	[−0.1672, −0.1025]	[−0.1557, −0.0617]	[0.0397, 0.0995]
largest p-value across refits	0.002	0.041	0.048
Matching without replacement	−0.0777	−0.0870	0.0644
	(0.0826)	(0.0795)	(0.0490)
p-value	0.375	0.306	0.225
Lead $k = -4$	−0.0446 [0.309]	−0.1057 [0.040]	0.1169 [0.002]
Lead $k = -3$	0.0377 [0.440]	−0.0161 [0.755]	0.0520 [0.214]
Lead $k = -2$	−0.0746 [0.078]	−0.0057 [0.834]	0.0509 [0.079]
joint p, pair-clustered	0.049	0.032	0.009
joint p, road bootstrap	0.348	0.482	0.221
tighter rule	0.295	0.031	0.395

Notes. Except for the without-replacement row, which is described below, all estimates use the baseline matched panel of 43 pairs on 8 comparison roads, with district-within-pair and pair-by-year fixed effects, clustered on the pair. The leave-one-comparison-road-out range covers the eight refits that drop each comparison road and every pair anchored on it. Matching without replacement solves an optimal one-to-one assignment on the same score, giving 9 pairs on 9 distinct roads and 1,005 observations. Lead coefficients are from the event study with $k = -1$ omitted, with p-values in brackets. The joint pre-period rows test $b(-4) = b(-3) = b(-2) = 0$, first with the analytic pair-clustered Wald test and then with a wild cluster bootstrap on the comparison road using Webb weights and 9,999 replications, on eight clusters under the baseline rule and six under the tighter one. *Source.* ENAHO 2004–2023 and Anexo 2 of Resolución 080-2021/CLC.

F.7 How far the contrast extends

The reported design puts every district within ten kilometres of a road into that road's catchment. That band is a choice, and the estimate it produces is a contrast between the districts immediately around a colluded road and those around its matched competitive road. This appendix asks how far the contrast reaches.

Everything is held at the reported baseline. The pairs are read from the baseline matching rather than rematched at each distance, and how many of them populate a given ring depends on the ring, and the tender dates, the four-year window, the two-year post lag, the district-within-pair and pair-by-year fixed effects are all unchanged, and every p-value quoted below is the comparison-road wild bootstrap. Only the distance band moves, in exclusive rings. The bands drawn in Figure 13 are the wider pair-clustered intervals, which are easier to read at this scale.

The area effect is not distinguishable from zero. Collapsing every district within forty kilometres into one observation per road-year gives an area effect of −0.008 on income, −0.004 on consumption and −0.011 on the poverty rate, none of them distinguishable from zero, under the fixed 2007 census population weight. The other two weights, surveyed households and surveyed persons, move the cells a little without making any of them distinguishable from zero, and the same holds whether the area outcome is a weighted mean of logs or a weighted mean of levels that is then logged. The census weight is time-invariant and predates most of the treatment, so it cannot respond to it. Whatever the ten-kilometre estimate is measuring, a fall of that size in average household welfare across the surrounding region is ruled out. A smaller one is not, since these estimates fail to reject zero rather than establish it. Table 17 reports the estimates with their intervals, which is what separates the two statements.

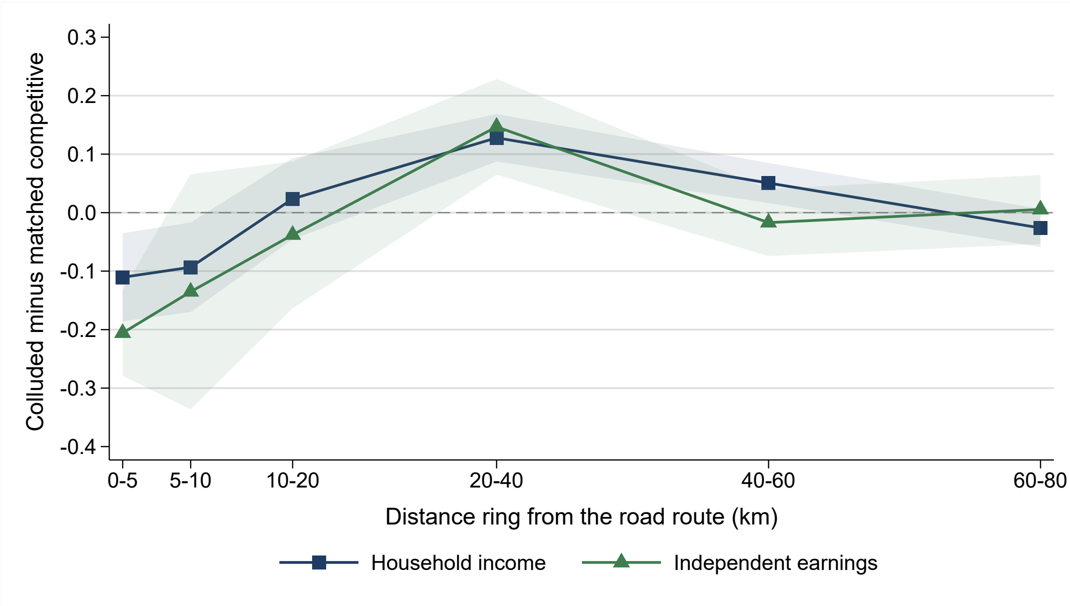
Table 17: The forty-kilometre area collapse under three weights.

	Surveyed households	Surveyed persons	Census 2007 population
Log income	-0.0049	-0.0081	-0.0080
	$p = 0.85$	$p = 0.72$	$p = 0.80$
95% CI	[-0.079, +0.067]	[-0.066, +0.052]	[-0.092, +0.054]
Log consumption	-0.0100	-0.0113	-0.0036
	$p = 0.78$	$p = 0.74$	$p = 0.93$
95% CI	[-0.105, +0.048]	[-0.095, +0.042]	[-0.111, +0.054]
Poverty rate	-0.0069	-0.0074	-0.0106
	$p = 0.74$	$p = 0.70$	$p = 0.62$
95% CI	[-0.049, +0.066]	[-0.054, +0.060]	[-0.057, +0.070]

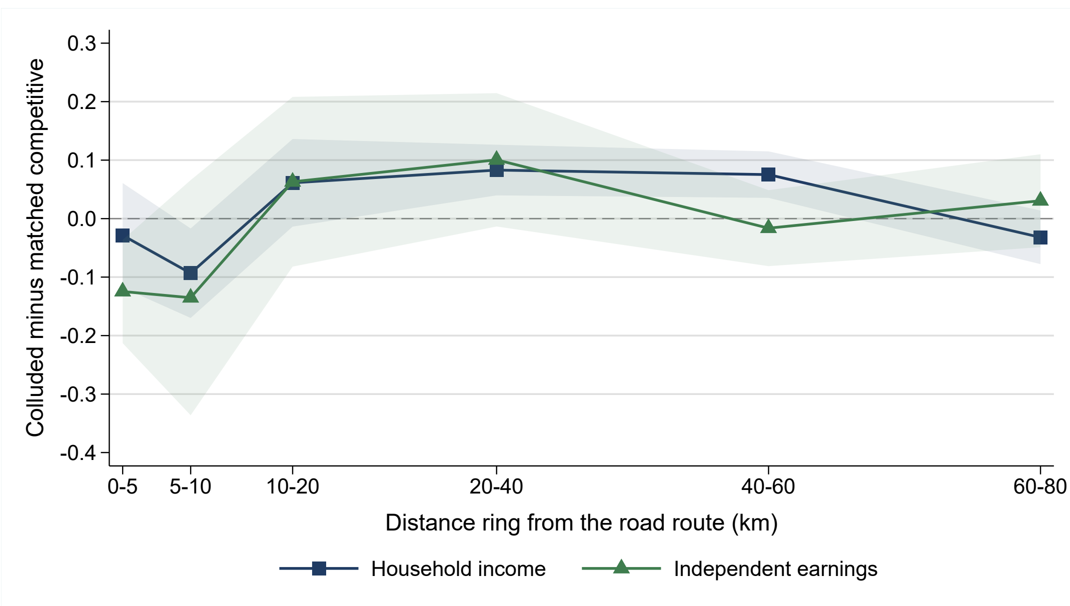
Notes. Every district within forty kilometres of a road is collapsed into one observation per road-year, so the unit is the whole local area rather than the district. The area outcome is a weighted mean of levels that is then logged, under the three weights named in the columns. Inference is the Webb-weight wild cluster bootstrap over the eight comparison roads, which is the level this design supports and the one the matched estimates are read at, with its p-value below each coefficient and its 95 percent interval beneath. Forty-three pairs and eight comparison roads. Clustering on the matched pair instead would roughly halve these intervals, the road-clustered standard error running up to 2.6 times the pair-clustered one, and it is not the inference the paper reads. The intervals are what the exclusion claim rests on. The most negative lower bound across income and consumption is -0.111, so an area effect the size of the ten-kilometre contrast, -0.168 on income and -0.160 on consumption under the tighter rule, lies outside every interval in the table. An area effect small enough to sit inside them does not, and the margin is not large. *Source.* ENAHO *sumaria* matched to Anexo 2.

The ring profile is suggestive and not stable. Figure 13 traces the estimate across six rings. Panel (a) uses every pair available in each ring and shows what looks like a clean spatial gradient, adverse against the route, crossing zero between ten and twenty kilometres, positive at twenty to forty, and back to zero by sixty to eighty. Panel (b) repeats it on the twenty pairs that have both arms populated in every ring, so that composition is held fixed and only distance moves. The near-road income coefficient falls from -0.111 to -0.029 and the positive band stops decaying where it did. Independent earnings keeps its shape better, negative against the route and positive at twenty to forty kilometres, but its innermost coefficient halves, from -0.206 to -0.125, and is no longer distinguishable from zero.

Part of that is power, since twenty pairs on six comparison roads is a much smaller design. But a coefficient moving from -0.111 to -0.029 is a change in magnitude and not in precision, so a material part of the apparent profile is which pairs enter which ring rather than distance itself. We therefore do not read the rings as evidence that activity relocated. The two claims we do make are that the contrast is concentrated immediately around the works and that aggregating the area does not produce a regional loss of the same size. A smaller one is not excluded.



(a) All pairs available in each ring



(b) The twenty pairs populated in every ring

Figure 13: The matched estimate across distance rings.

Notes. Each point is the matched road-package estimate computed on an exclusive distance ring, with the same pairs, tender dates, window, lag and fixed effects as the reported design. Shaded bands are 95 percent intervals clustered on the matched pair. Panel (a) uses whichever pairs have both arms populated in a given ring, between 20 and 45 of the 47 candidate pairs the baseline matching produces, 43 of which populate the reported ten-kilometre design. Panel (b) fixes the sample to the 20 pairs populated in all six rings, on 6 comparison roads, so the profile cannot reflect composition. *Source.* ENAHO *sumaria* and the employment and income module, matched to Anexo 2.

G Raw cartel event studies and poverty

Figure 14 reports the cartel-exposure event studies for income and consumption without the pre-period normalization used in the main text, each estimator on its own native baseline. Callaway–Sant'Anna and de Chaisemartin–D'Haultfœuille have flat raw pre-trends and an adverse post-treatment path, two-way fixed effects corroborates the sign, and the Borusyak estimator has noisier raw pre-period point estimates and is sensitive to the normalization, which is why the main text relies on the two staggered-robust estimators.

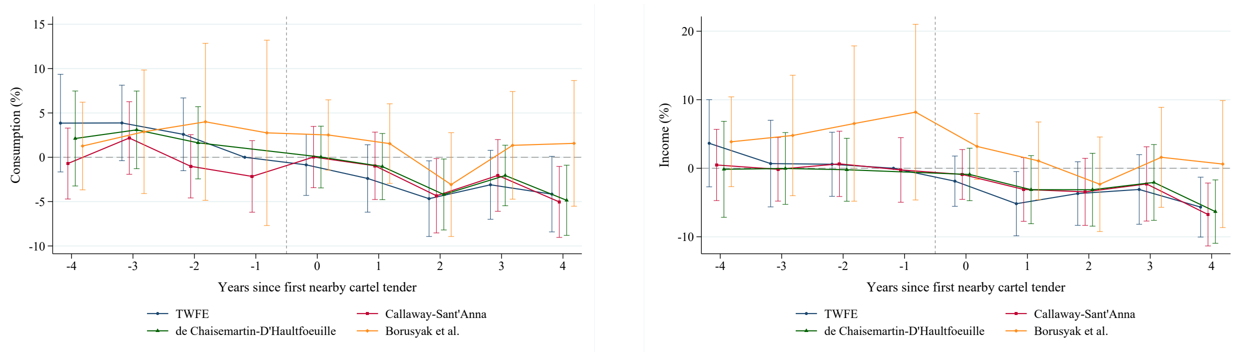


Figure 14: Raw, uncentered cartel event studies: consumption (left) and income (right).
Notes. Coefficients shown without pre-period normalization. *Source.* ENAHO *sumaria* district panel and Resolución 080-2021/CLC.

Figure 15 reports the poverty event study, which is directionally adverse but weaker than consumption and income.

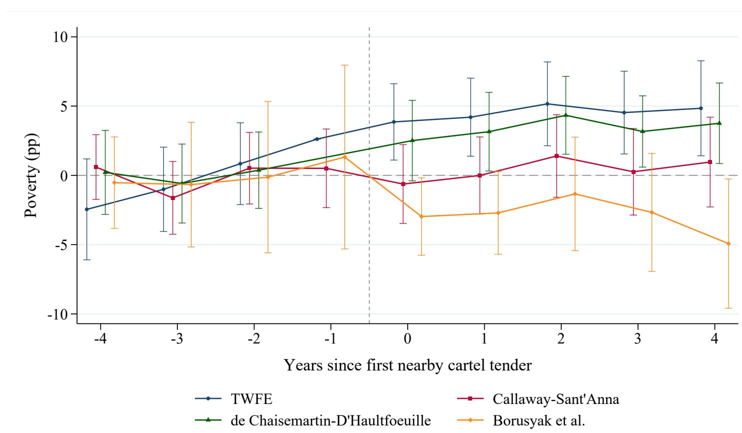


Figure 15: Poverty-rate event study by four estimators, cartel exposure.
Notes. Normalized to each estimator's pre-period mean. *Source.* ENAHO *sumaria* district panel and Resolución 080-2021/CLC.

H The antitrust resolution, reconstruction and cartel governance

This appendix documents what Section 4.2 summarizes: the resolution, the reconstruction of the process universe from the firm exhibits, the governance and enforcement evidence, and the validation against the procurement records.

H.1 The resolution

Indecopi, the Peruvian competition authority, sanctioned the cartel in Resolución 080-2021/CLC-INDECOPI of 15 November 2021 (Expediente 001-2020/CLC). The procedure had opened with the Resolución de Inicio 003-2020/ST-CLC of 31 January 2020, and the imputed firms were notified in early February 2020. The resolution declares that thirty-two economic groups engaged in a horizontal bid-rigging agreement, coordinating cover bids and abstentions in the public procurement of road works at the national level between November 2002 and December 2016, and it fines them in units of the *Unidad Impositiva Tributaria* (UIT). The conduct periods are firm specific. Sixteen groups participate from the cartel's start in November 2002, and sixteen enter later, between 2004 and 2014. The resolution is the authoritative source for who was in the cartel, for each firm's role in each tender, and for the award outcomes. We take winners, members, and roles from it throughout.

H.2 The process universe

The resolution documents the cartel's allocation through firm-specific evidence exhibits, in which each firm's own records list the tenders it participated in and the consortium that took each award. Appendix A documents the full reconstruction and validation of both judicial datasets, the Construction Club cartel here and the Lava Jato concessions of Section 6.1. We read three structured exhibits directly from the resolution at the page level, the Cosapi exhibit (CSP18, twenty-five processes), the Obrainsa exhibit (OBN10, thirty-eight processes), and the JJC exhibit (JJC01, fifty-two entries over forty-nine distinct processes). The remaining named evidence (Johesa, GyM, San Martín and others) is narrative and not a process table, and we use it only to corroborate participation, not to add processes.

Merging the three exhibits requires care because the same tender appears under different code notations across firms, for example a *clásico* number in one exhibit and an MTC code in another. We match on a conservative full code signature and confirm each candidate on project name and winner before merging, so that distinct sub-tramos and distinct lots are never collapsed by code similarity alone. The union is *seventy-eight distinct processes* spanning 2002 to 2016. For each process we record the designated winning consortium, its member firms, and the value figures the resolution reports. These seventy-eight are the processes we reconstruct in firm-level detail from the exhibits, and they are narrower than the resolution's own annexes that the analysis below also draws on. Anexo 2 carries the adjudicated classification over the 206 procedures it covers, of which 145 carry bids a screen can be computed on, and that subsample is the basis for the median price comparison and the detection screens of Section 6.3. The controlled price regression of Table 6 runs on a differently filtered 122 tenders that the note to that table sets against these 145, while Anexo 3 reports the competition authority's illicit-benefit and penalty computation over the full sanctioned caseload, which is the basis for the aggregate overcharge in Section 6.2. A figure in what follows refers either to these reconstructed processes or to one of those annexes, and we make clear which. For each firm in each process we record a role, winning-consortium member, cover bidder, cover bidder of uncertain target (the resolution's category for support whose beneficiary the firm did not recall), or participant without support.

H.3 Governance and enforcement

Figure 16 sketches how the cartel operated, from coordination to the overcharge.

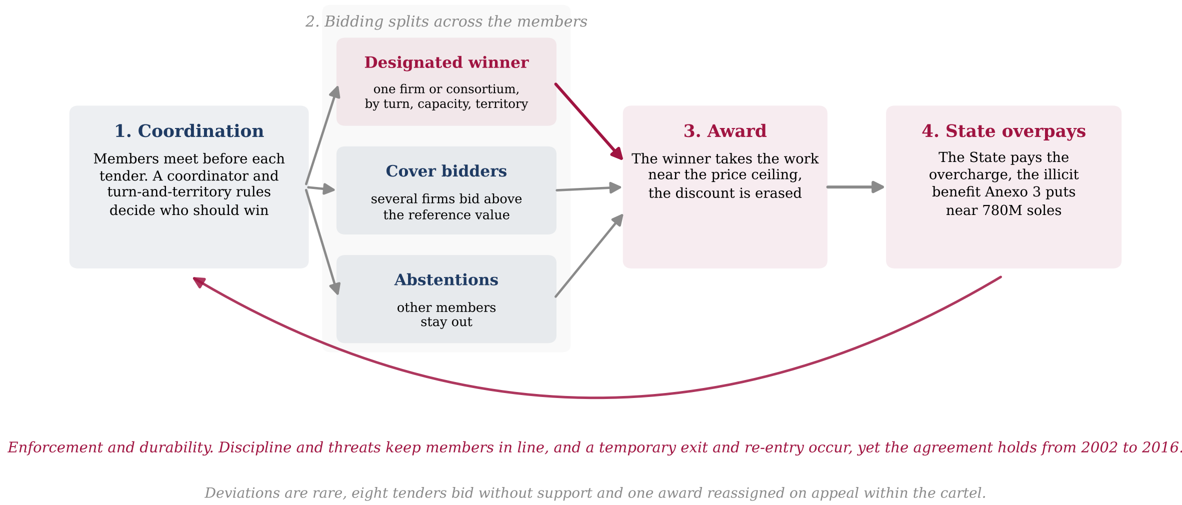


Figure 16: How the Construction Club cartel operated.

Notes. Members coordinated before each tender, designated a winner, one firm or a consortium of them, by turn, capacity and territory, and the others submitted cover bids above the reference value or abstained, so the winner took the work near the price ceiling and the State paid the overcharge. Discipline and a documented exit and re-entry kept the agreement in force from 2002 to 2016. *Source.* Resolución 080-2021/CLC.

The resolution records who won, who covered, and how the cartel governed itself, and this matters for the mechanism. A mechanical rotation table would leave little room for a sustained overcharge, but the evidence describes a durable institution that managed entry, assignment, and discipline over fourteen years. Table 18 collects the documented episodes. The assignment of each process was preceded by coordination meetings, general and one-on-one, in which firms declared their interest in particular works, and a standing coordinator organized the smaller firms. The arrangement survived friction. When one of the largest members withdrew in 2007, the remaining firms continued to coordinate and then arranged its reincorporation, which the competition authority reads as the persistence of the agreement and not its collapse. Discipline was explicit, an internal email records a signed side-agreement and the understanding that a member who broke ranks would cease to belong. Deviations occur in the record, eight tenders in which a cartel firm bid without supporting the designated winner, and one award that passed on appeal from the disqualified designate to another cartel member, but they are deviations within a functioning system, not its breakdown.

Table 18: Cartel governance, deviations, and enforcement.

Episode type	Period	What the resolution documents	Evidence
Coordination role	2002–2016	Grupo Plaza acted as the standing coordinator of the small and medium firms, named in the declarations of Cosapi and JJC executives.	Aranda, Martínez (p. 362)
Pre-assignment meetings	2002–2016	The assignment of each process was preceded by meetings in which members declared their interest in a given work, held in named restaurants and offices, both general and one-on-one.	Testimony (p. 158, 283)
Temporary exit and re-entry	2007–2008	JJC withdrew in 2007 while the remaining members kept coordinating, and other firms then coordinated JJC's reincorporation into the agreement.	GyM18, GyM20 (p. 362)
Signed side-agreement and enforcement	2011	An internal email records a signed agreement with an outside representative and the threat that a member who broke the agreement would cease to be part of the arrangement.	CSP13 (p. 1387)
Documented deviations	2002–2016	Eight tenders in which a cartel firm participated without supporting the designated winner, the resolution's no-support category.	JJC01 (p. 312–313)
Reassignment on appeal	2010	The designated winner of one tender was disqualified and the award passed on appeal to another cartel member, the coordination rerouting rather than failing.	CSP18 (p. 315)

Notes. Episodes coded from the narrative and the evidence exhibits of Resolución 080-2021/CLC, with the resolution's own evidence codes and page references. The episodes describe how the cartel was governed rather than how often it failed, and the competition authority finds that the coordination continued across the deviations and the temporary exit. *Source.* Resolución 080-2021/CLC.

The allocation the governance produced looks like division by turns more than dominance. Documented wins are diffuse, with a winner Herfindahl of 0.06 and the top five groups taking under half the winning positions, consistent with a sharing arrangement in which members take turns. Part of that diffuseness is joint bidding rather than rotation, since a win is membership in the winning consortium and more than half the awarded tenders went to a consortium of two to four firms. Two regularities shape who takes which turn. Allocation tracks capacity, since participation and wins correlate at 0.72, and it tilts by territory, since groups with three or more wins concentrate 61 percent of them in a single geographic zone, the market-division pattern that Marshall and Marx (2012) describe. This governance is what makes a local effect conceivable. A cartel that endures, meets, assigns, and enforces can hold the winning price near the ceiling the State will pay, year after year, on the works that build a district's roads. The price of that durability is the overcharge documented in Section 6.2. Whether it reached the households around those works is the question

Section 7.2 takes up, and the governance evidence is why the question is worth asking.

H.4 External validation

The universe is validated against Peru's open procurement data in Appendix C, which recovers seventy-three of the seventy-eight processes and matches sixty-three of them at high confidence. The procurement data confirms that these are real tenders of the right entity, year, work, and order of magnitude.

Two limits of the procurement data shape how we use it. First, *SEACE is not a usable source of winners* for the large road licitaciones. The award and supplier records are empty for the big works, so the winner-to-firm link is absent for exactly the processes that matter. Winner identity therefore comes from the resolution, and SEACE serves only to validate existence, code, entity, work, date, and the order of magnitude of the value. Second, for the same reason we *do not construct a cartelized-share-of-procurement measure*. A firm-level share would divide cartelized value by the firm's total public-contract value, but the denominator from SEACE awards is too incomplete and uneven for the cartel firms, fourteen of the thirty-two have no award records at all and the large-works winners appear in only a handful, so the share would be biased upward for the most relevant firms. We use a measure built entirely from the judicial record instead.

I The overrun comparison in detail

Section 6.1 reports the overrun comparison across the three documentation categories. Two further displays are collected here. The first is the binary specification, which pools the judicially implicated projects with those carrying no documented evidence and therefore measures the gap against a mixed comparison group. The second is the small-sample inference battery, which subjects the unconditional gap to randomization inference over the bribe labels, a Webb-weight wild bootstrap, a rank-sum test and a leave-one-megaproject-out range, and records that the conditional gap weakens under an HC3 standard error.

Table 19: Documented bribery and cost overruns (descriptive, $N = 25$).

	(1) bivariate	(2) +controls	(3) bribe or implication +controls
Treatment indicator	36.0***	22.1*	20.3*
	(12.7)	(11.2)	(10.3)
Transport		20.8	26.9*
		(13.8)	(14.8)
log initial investment		11.7**	12.0**
		(4.8)	(4.7)
N projects	25	25	25
R^2	0.281	0.482	0.489

Notes. Cross-sectional regression of the cost overrun (percentage points of initial investment) on the documented-bribe indicator, $N = 25$ Odebrecht projects in Peru. Columns (1) and (2) compare the ten documented-bribe projects against the fifteen not documented as bribed. Column (3) widens the treatment to the fifteen projects carrying either a documented bribe or a judicial implication, so its comparison group is the ten projects with no documented evidence. Columns (2) and (3) also include a linear year term, which is not displayed. Dropping it moves the column (2) estimate from +22.1 to +33.8, so it is a control and not a nuisance. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *Source.* Odebrecht–Peru case data, Campos et al. (2021).

Table 20: Small-sample robustness of the overrun gap ($N = 25$).

Test	Result
Baseline gap (documented bribe vs not documented as bribed)	+36 pp
With sector fixed effects, size and year	+25.1 pp (HC1 SE 11.8)
Randomization inference over bribe labels	$p = 0.005$
Webb-weight wild-cluster bootstrap	$p = 0.005$, CI $[10, 62]$
Wilcoxon rank-sum test	$p = 0.002$
Leave-one-megaproject-out range	$[25, 40]$ pp
Conditional gap, HC3 standard error	SE 15.97, $p = 0.116$
Monotone three-bucket overrun means (%)	10.5/28.6/52.6

Notes. Robustness of the cost-overrun gap between the ten documented-bribe Odebrecht projects and the fifteen not documented as bribed, $N = 25$. The baseline +36 pp gap is the bivariate regression in column (1) of Table 19. The conditional gap reported here regresses the overrun on the documented-bribe indicator with a full set of sector fixed effects, log initial investment and a linear year term, contrasting the ten documented-bribe projects with the other fifteen pooled. It is +25.1 pp with an HC1 standard error of 11.8 ($p = 0.034$) and an HC3 standard error of 15.97 ($p = 0.116$). It is not the +22.1 pp of column (2) of Table 19, where a single transport indicator replaces the sector fixed effects, and it is not the +28.4 pp of column (3) of Table 4, which shares this control set but keeps the judicially implicated projects in a separate category instead of pooling them with the no-evidence group. The three-bucket means are the average overruns for the ten no-evidence, five judicially-implicated, and ten confirmed-bribe projects, so the +36 pp baseline compares the last group against the first two pooled. With twenty-five projects the gap is a description, not an estimated elasticity, and the conditional gap weakens honestly under HC3. *Source.* Odebrecht–Peru case data, Campos et al. (2021).

J Additional household tables

Three tables supporting Section 7.2 are collected here. The first places the corridor and cartel exposure measures in the same district panel, where cartel exposure is adverse, corridor exposure is not distinguishable from zero, and the interaction of the two is adverse, so the districts reached by both fare worst. The second is the colluded-versus-competitive comparison across the geographic fixed-effect ladder, whose sign is adverse throughout but whose common support is too thin to separate corruption from the road. The third reports the matched road-package regression under the baseline rule with its full ladder of inference levels, the baseline coefficients of which also appear in Table 9.

Table 21: Lava Jato corridor exposure versus Construction Club cartel exposure.

	(1) Lava Jato corridor only	(2) Construction Club cartel only	(3) Both branches	(4) Both + interaction
Log household income				
Lava Jato corridor exposure	−0.016		−0.016	−0.009
	(0.018)		(0.018)	(0.018)
Construction Club cartel exposure		−0.016**	−0.016**	−0.011*
		(0.007)	(0.007)	(0.006)
Lava Jato × Construction Club				−0.027***
				(0.005)
Log household consumption				
Lava Jato corridor exposure	−0.019		−0.018	−0.012
	(0.016)		(0.016)	(0.016)
Construction Club cartel exposure		−0.017***	−0.016***	−0.012**
		(0.005)	(0.005)	(0.005)
Lava Jato × Construction Club				−0.023***
				(0.005)
Poverty rate				
Lava Jato corridor exposure	0.015		0.015	0.010
	(0.014)		(0.014)	(0.014)
Construction Club cartel exposure		0.010	0.009	0.006
		(0.006)	(0.006)	(0.005)
Lava Jato × Construction Club				0.016***
				(0.004)
District FE	Yes	Yes	Yes	Yes
Region × year FE	Yes	Yes	Yes	Yes
Observations	21,900	21,900	21,900	21,900

Notes. Coefficients are post-treatment effects per one standard deviation of exposure. Lava Jato corridor exposure is the concession-corridor market-access measure on the corridor opening clock. Construction Club cartel exposure is the proximity-decayed district exposure to the 76 geocoded Indecopi processes on the first-nearby-process clock, whose post period opens two years after that first process. All columns include district and region-by-year fixed effects, with standard errors clustered by region. Stars denote $p < 0.1$, $p < 0.05$, $p < 0.01$.

Table 22: Colluded versus competitive road exposure, by geographic fixed effect.

	Colluded road exposure	Competitive road exposure	Difference $\delta^C - \delta^K$
Log household income			
District + department×year	−0.027	0.009	−0.036
	(0.028) [0.036]	(0.620)	(0.024), 0.14 [0.17]
District + province×year	−0.023	0.013	−0.036
	(0.165)	(0.649)	(0.035), 0.31 [0.37]
within both-arm provinces	−0.035	0.018	−0.053
	(0.474)	(0.331)	(0.058), 0.39 [0.47]
+ road-size control	−0.032	0.008	−0.040
	(0.066) [0.100]	(0.684)	(0.025), 0.11 [0.15]
Log household consumption			
District + department×year	−0.024	−0.004	−0.020
	(0.024) [0.037]	(0.805)	(0.020), 0.32 [0.36]
District + province×year	−0.018	0.002	−0.019
	(0.220)	(0.935)	(0.027), 0.48
within both-arm provinces	−0.049	0.019	−0.068
	(0.153)	(0.021)	(0.030), 0.07 [0.13]
+ road-size control	−0.045	−0.009	−0.035
	(0.002) [0.008]	(0.533)	(0.021), 0.09 [0.14]
Poverty rate			
District + department×year	0.013	0.007	0.006
	(0.033) [0.044]	(0.310)	(0.010), 0.60 [0.58]
District + province×year	0.010	0.001	0.009
	(0.229)	(0.927)	(0.015), 0.53 [0.55]
within both-arm provinces	0.021	0.003	0.019
	(0.264)	(0.826)	(0.024), 0.47 [0.44]
+ road-size control	0.027	0.011	0.016
	(0.004) [0.022]	(0.146)	(0.011), 0.15 [0.17]
Observations (department, province rows)		4,659	
Observations (within both-arm provinces)		414	

Notes. Coefficients are post-treatment effects per one standard deviation of proximity-decayed exposure, from a single regression entering colluded and competitive road exposure together. Colluded and competitive tenders are classified by the tender-level *tipo de proceso* column of Anexo 2 of Resolución 080-2021/CLC, which is constant across the offers of a tender and is the label the matched design and every household regression use; the offer-level *tipo de oferta* column is carried in the data but never defines treatment and are bid by the same cartel firms on the same class of national road works. Each panel adds a finer geographic control, from department-by-year to province-by-year to a within-province comparison restricted to the eight provinces that contain both a colluded and a competitive tender (27 districts). The road-size control adds the reference-value-weighted total of nearby road exposure at the department-by-year level. In the first two columns, analytic p-values clustered by province are in parentheses and wild-cluster bootstrap p-values in brackets where computed. The third column reports the linear combination $\delta^C - \delta^K$, which Section 5 defines as the corruption-specific component net of the road, with its standard error in parentheses followed by the analytic and, in brackets, the wild-cluster bootstrap p-value. The wild-cluster p is omitted for the province-by-year consumption row, where the bootstrap variance matrix is singular. The difference is adverse for income and consumption on every rung but is not distinguishable from zero at conventional levels anywhere, including the two cells that approach them analytically, consumption within the both-arm provinces and consumption with the road-size control, both of which move away from significance under the small-cluster bootstrap. The 95 percent interval for the department-level income difference is $[-0.085, 0.013]$. Significance of δ^C against zero is therefore not evidence that δ^C differs from δ^K. *Source.* ENAHO *sumaria* district panel matched to Anexo 2.

Table 23: Matched road-package stacked difference-in-differences.

	(1) Consumption	(2) Income	(3) Poverty
Colluded × Post	-0.114^{*}	-0.144^{**}	$+0.068^{*}$
	(0.049)	(0.052)	(0.033)
District-within-pair FE	Yes	Yes	Yes
Pair × year FE	Yes	Yes	Yes
Observations	4,257	4,257	4,257
District clusters	301	301	301
p-value by level of inference			
District clusters (301)	0.019	0.006	0.042
Matched pairs (43)	0.001	0.000	0.002
Competitive comparison roads (8)	0.111	0.026	0.128
Wild bootstrap, comparison roads (8)	0.179	0.061	0.180
Matched-pair permutation	0.062	0.028	0.067

Notes. Post-period effect, event years two to four, of exposure to a colluded road package relative to its matched competitive package, on a four-year window around the tender, with district-within-pair and pair-by-year fixed effects and standard errors clustered by district. Forty-three matched pairs built from eight competitive comparison roads, reused unevenly, with one road anchoring twelve pairs. The lower panel holds the coefficients fixed and varies only the level of inference. The competitive-comparison-road rows cluster on those eight roads, the wild bootstrap is a Webb-weight cluster bootstrap over the same eight, and the permutation test flips the colluded and competitive labels within each matched pair over 1,999 draws, comparing studentized statistics. Stars denote significance under the district clustering of the top panel and are not carried by the package-level inference. *Source.* ENAHO *sumaria* matched to Anexo 2.

K Public investment and crowd-out

A natural alternative to a household-incidence reading is that the cartel's overcharge crowded out other local public investment. Using the Invierte.pe project registry we measure public investment by function for districts and regions more exposed to the colluded tenders. Non-transport investment does not fall after exposure. The effect on non-road investment is indistinguishable from zero at the district level, and while negative in point estimate at the region level it is imprecise and not echoed by the non-road share, so it does not survive as displacement. At the national level transport and non-transport investment rose together. The data do not favour a local reallocation account. They do not establish where the fiscal cost lands either, since a displacement too small for this design to detect would leave the same trace.

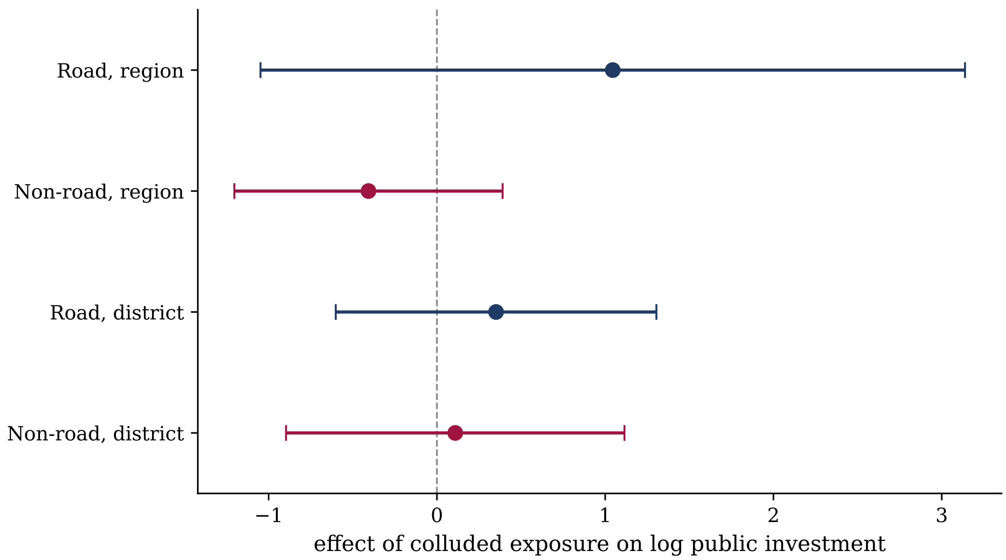


Figure 17: Fiscal crowd-out: effect of colluded-cartel exposure on public investment, by function and margin.

Notes. Each point is the coefficient on colluded-cartel exposure interacted with post-exposure, in a panel of public investment (Invierte.pe initiation-year project cost) with two-way fixed effects, district and region-by-year at the district margin and region and year at the region margin, clustered at the coarser unit. Bars are 95 percent confidence intervals. Non-road investment (the crowd-out outcome) is indistinguishable from zero at both margins, while road investment carries a positive point estimate that confirms the exposure tracks road activity. *Source.* MEF Invierte.pe project registry and Resolución 080-2021/CLC.

L Additional figures and the full firm table

Table 24: Cartel allocation and contract capture, all firms appearing in the reconstructed process universe.

Firm	Won proc.	Cover proc.	No-supp. / unc.	Ref. value won (S/. M)	Departments	Indep. evid.
JJC	14	28	8	1,111	Amazonas, Ancash, Apurimac +17	27
Iccgsa	11	0	0	1,102	Amazonas, Apurimac, Ayacucho +6	7
CASA	7	0	0	757	Arequipa, Cajamarca, Cusco +6	4
Johesa	7	0	0	713	Ayacucho, Cajamarca, Huancavelica +3	2
E. Reyna	5	0	0	553	Arequipa, Cajamarca, Lambayeque +4	1
Energoprojekt	5	0	0	816	Cajamarca, Cusco, Huancavelica +3	3
Obrainsa	4	38	0	–	Amazonas, Ancash, Apurimac +16	18
Constructora Malaga	4	0	0	933	Apurimac, Ayacucho, Lima +3	3
Queiroz Galvao	4	0	0	559	Cajamarca, Cusco, Huanuco +2	3
Altesa	3	0	0	23	Ancash, Cajamarca, Cusco +1	2
Aramsa	3	0	0	406	Huanuco, Pasco, Ucayali	1
Camargo Correa	3	0	0	374	Cajamarca, Huanuco, Lambayeque +1	2
Conalvias	3	0	0	496	Cajamarca, Cusco, Lambayeque +1	2
Grupo OHL	3	0	0	600	Apurimac, Ayacucho, Cusco	1
GyM	3	0	0	571	Apurimac, Ayacucho, Huancavelica	2
Upaca	3	0	0	384	Ancash, Huancavelica, Lima	1
Cosapi	2	25	0	409	Apurimac, Arequipa, Ayacucho +11	19
Aterpa	2	0	0	71	Amazonas, Lima	0
Conciviles	2	0	0	329	Cajamarca, Cusco, Lambayeque	2
Constructora TP	2	0	0	490	Apurimac, Ayacucho, Cusco	1
JCCG	2	0	0	182	Apurimac, Ayacucho	0
Superconcreto	2	0	0	404	Cajamarca, La Libertad, Lambayeque +1	1
Andrade Gutierrez	1	0	0	469	Lima	1
CyM	1	0	0	110	Ayacucho, Huancavelica	0
Eivisac	1	0	0	442	Ayacucho	1
Grupo Plaza	1	0	0	163	La Libertad	1
OAS	1	0	0	–	Lima	1
San Martin	1	0	0	357	Pasco, Ucayali	0

Notes. The table reports contract allocation and capture within the Construction Club, reconstructed from Resolución 080-2021/CLC. It does not measure firm profits. Won and cover-bidder columns count processes in which the firm was a member of the winning consortium or a documented cover bidder. Reference value won sums available reference values of won processes and is a lower bound given partial value coverage. Independent evidence counts processes corroborated by more than one firm exhibit.

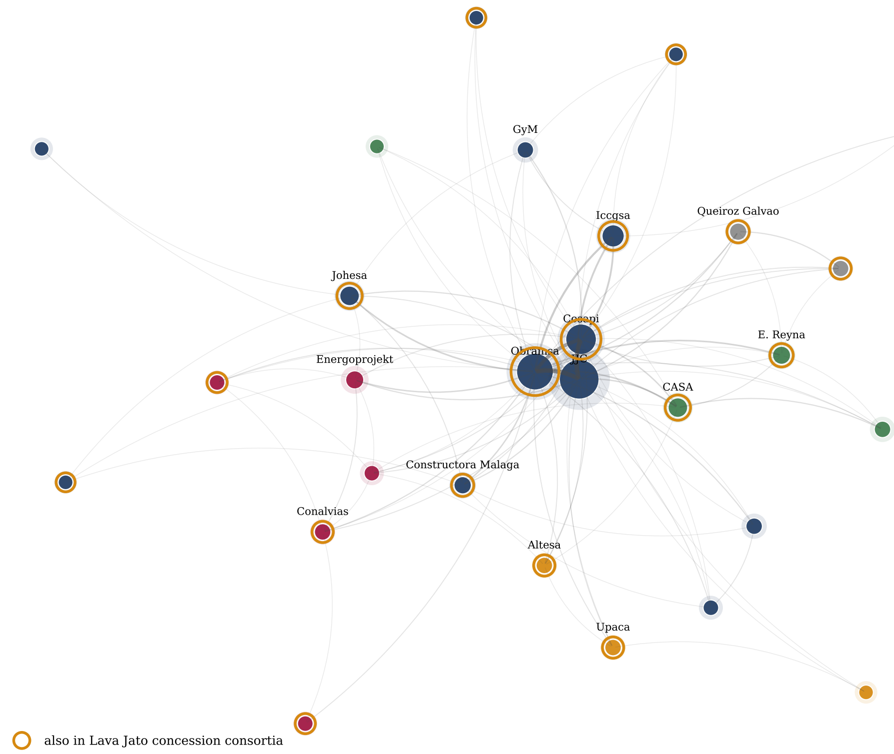


Figure 18: Cartel co-participation network.

Notes. Firms that bid in the same cartelized tender are linked, with edge width and shade the number of shared tenders. Node size is the firm's documented involvement and color marks the sub-communities a modularity partition finds. The dense center is the cartel core, partly reflecting the three firms with full bid exhibits (Cosapi, Obrainsa, JJC), around which territorial and turn-taking sub-groups sit. Gold rings mark the cartel groups in this network that also appear in the Lava Jato concession consortia. *Source.* Resolución 080-2021/CLC.

The firm-role matrix in Figure 19 gives the same allocation as counts.

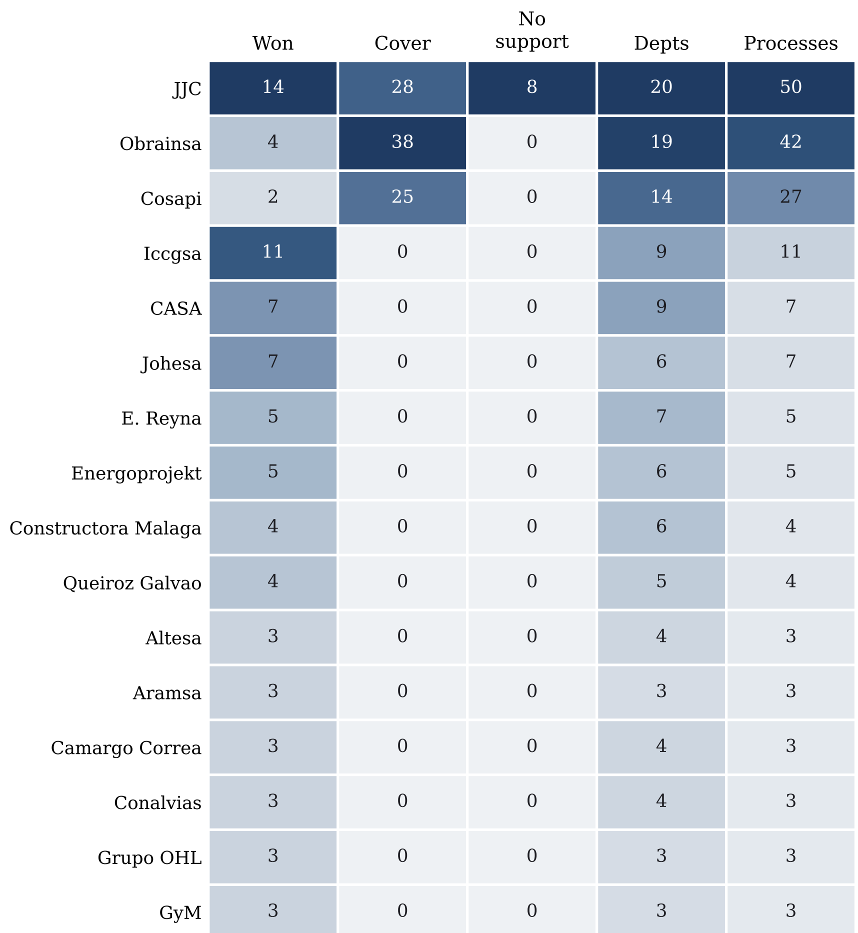
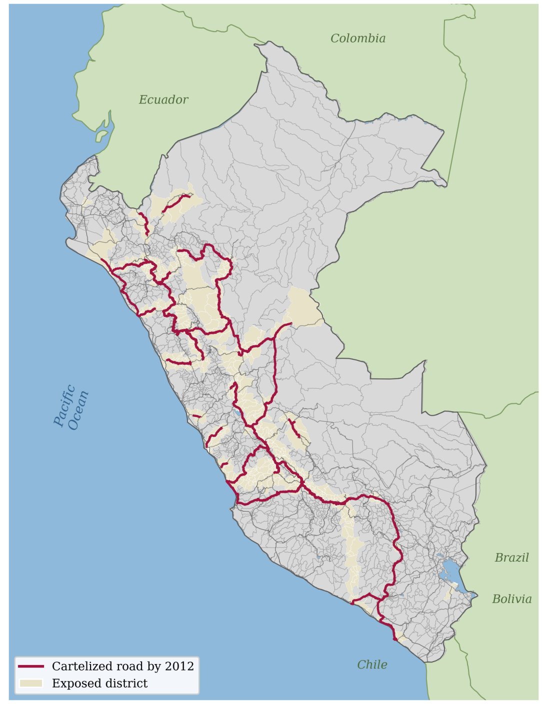


Figure 19: Firm-by-role matrix: won, cover, no-support, departments, processes.
Notes. Each column is colour-normalized to its own maximum, with the raw count shown. Firms ordered by total documented involvement. *Source.* Resolución 080-2021/CLC.

The cartel footprint accumulates over time as more tenders are let, shown in Figure 20.

(a) Reached by 2008



(b) Reached by 2012

(c) Reached by 2016

Figure 20: Evolution of cartel exposure: districts and cartelized routes reached by 2008, 2012, and 2016.

Notes. Cumulative districts within twenty kilometers of a cartel process let by the year shown, with the cartelized routes let by that year. *Source.* Resolución 080-2021/CLC, MTC road network.